



Asymptotic stability for a strongly coupled Klein-Gordon system in an inhomogeneous medium with locally distributed damping

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Dedicated to Professor Jaime E. Muñoz Rivera on the occasion of his 60th Birthday

Abstract

We consider a strongly coupled Klein-Gordon system posed in an inhomogeneous medium Ω with smooth boundary $\partial\Omega$ subject to a local damping distributed around a neighborhood ω of the boundary according to the Geometric Control Condition. We show that the energy of the system goes uniformly and exponentially to zero for all initial data of finite energy taken in bounded sets of finite energy phase-space. For this purpose, refined microlocal analysis arguments are considered by exploiting ideas due to Burq and Gérard [16]. By using sharp $L^p - L^q$ -Carleman estimates we prove a unique continuation property for coupled systems. We also consider a strongly coupled system of wave equations with localized nonlinear damping acting only on one equation. Considering a compact Riemannian manifold, we show that the energy of the coupled system goes uniformly to zero.

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1. Introduction

1.1. Description of the problem

This article addresses the exponential stability of a semilinear wave equation system posed in an inhomogeneous medium and subject to a frictional damping locally distributed:

$$\begin{cases} \rho(x)u_{tt} - \operatorname{div}(K(x)\nabla u) + v^2u + a(x)u_t - \gamma(x)v_t = 0 & \text{in } \Omega \times (0, +\infty), \\ \rho(x)v_{tt} - \operatorname{div}(K(x)\nabla v) + u^2v + b(x)v_t + \gamma(x)u_t = 0 & \text{in } \Omega \times (0, +\infty), \\ u = v = 0 & \text{on } \partial\Omega \times (0, +\infty), \\ u(x, 0) = u_0(x), \quad u_t(x, 0) = u_1(x) & \text{in } \Omega, \\ v(x, 0) = v_0(x), \quad v_t(x, 0) = v_1(x) & \text{in } \Omega, \end{cases} \quad (1.1)$$

where $\Omega \subset \mathbb{R}^d$, $d = 2, 3$, is a bounded domain with a sufficiently smooth boundary $\Gamma = \partial\Omega$, $\rho : \Omega \rightarrow \mathbb{R}_+$, $k_{ij} : \Omega \rightarrow \mathbb{R}$, $1 \leq i, j \leq d$ are $C^\infty(\Omega)$ functions such that for all $x \in \Omega$ and $\xi \in \mathbb{R}^d$,

$$\alpha_0 \leq \rho(x) \leq \beta_0, \quad k_{ij}(x) = k_{ji}(x), \quad \alpha|\xi|^2 \leq \xi^\top \cdot K(x) \cdot \xi \leq \beta|\xi|^2, \quad (1.2)$$

where $\alpha_0, \beta_0, \alpha, \beta$ are positive constants and $K(x) = (k_{ij})_{i,j}$ is a symmetric positive-definite matrix. We denote by ω the intersection of Ω with a neighborhood of $\partial\Omega$ in $\mathbb{R}^d$.

Assumption 1.1. The non-negative functions $a(\cdot)$ and $b(\cdot)$, responsible for the localized dissipative effect, satisfies the following conditions:

$$a(\cdot), b(\cdot) \in C^0(\overline{\Omega}) \text{ with } a(x) \geq a_0 > 0 \text{ in } \omega \subset \Omega \text{ and } b(x) \geq b_0 > 0 \text{ in } \omega \subset \Omega. \quad (1.3)$$

In addition, let us assume that $\gamma \in W^{1,\infty}(\Omega)$ and

$$0 \leq \gamma(x) \leq a(x) \text{ and } 0 \leq \gamma(x) \leq b(x) \text{ a. e. in } \Omega. \quad (1.4)$$

Let us assume that the following assumptions are also made:

Assumption 1.2. ω geometrically controls Ω , i.e. there exists $T_0 > 0$, such that every geodesic of the metric $G(x)$, where $G(x) = \left(\frac{K(x)}{\rho(x)}\right)^{-1}$ traveling with speed 1 and issued at $t = 0$, intercepts ω in a time $t < T_0$.

Assumption 1.3. For every $T > 0$, the only solution $u, v \in C([0, T[; L^2(\Omega)) \cap C([0, T[, H^{-1}(\Omega))$ to the system

$$\begin{cases} \rho(x)u_{tt} - \operatorname{div}(K(x)\nabla u) + V_1(x, t)u = V_3(x, t)v & \text{in } \Omega \times (0, T), \\ \rho(x)v_{tt} - \operatorname{div}[K(x)\nabla v] + V_2(x, t)v = V_3(x, t)u & \text{in } \Omega \times (0, T), \\ u = v = 0 & \text{on } \omega, \end{cases} \quad (1.5)$$

where $V_1(x, t)$, $V_2(x, t)$ and $V_3(x, t)$ are elements of $L^\infty([0, T[, L^{\frac{d+1}{2}}(\Omega))$, is the trivial one $u = v = 0$.

Assumption 1.2 is the so called Geometric Control Condition (G.C.C.). It is well-known that it is necessary and sufficient for stabilization and control of the linear wave equation, see [11], [15], [19], [21], [28], [48] and the references therein. In this direction, it is worth mentioning the work due to Betelú, Gulliver and Littman [12] where the authors discussed the question of **closed geodesics** in the interior of a region Ω . Such closed geodesics make control impossible since they are bicharacteristics that never reach the controlled boundary. In this paper, they show that, in the two-dimensional case, the lack of closed geodesics in the interior is also sufficient for control. For this reason and since in the present paper we do not have any control of the geodesics because of the inhomogeneous medium we are forced to consider ω a neighborhood of the whole boundary $\partial\Omega$ and to give conditions on the metric $G = (K/\rho)^{-1}$ so that every geodesic of the metric G enters the set ω in a time $t < T_0$. In a general setting, we must assume that $a, b \in C^0(\overline{\Omega})$ and the following assumptions must be considered:

- (i) $\forall x \in \partial\Omega, a(x), b(x) > 0$,
- (ii) For all geodesic $t \in I \mapsto x(t) \in \Omega$ of the metric $G = (K/\rho)^{-1}$, with $0 \in I$, there exists $t \geq 0$ such that $a(x(t)), b(x(t)) > 0$.

We observe that if $G = I_d$ the condition (i) above implies that (ii) holds since the geodesics are straight lines. On the other hand, for a general metric condition, (i) can be verified even if (ii) does not hold, if, for instance, G admits a trapped geodesic (see Fig. 1).

Remark 1. It is important to observe that Assumption 1.2 is not obviously fulfilled for every matrix $G = (K/\rho)^{-1}$. For instance, the equator in the unit sphere S^2 is a periodic geodesic and we can consider $\Omega \subset S^2$ that contains the equator as a domain in $\mathbb{R}^2$ endowed with a Riemannian metric G . We shall give in the appendix of this manuscript examples where this situation occurs. An easy one happens when $G(x) = I_d$. In this case the geodesics are straight lines and necessarily

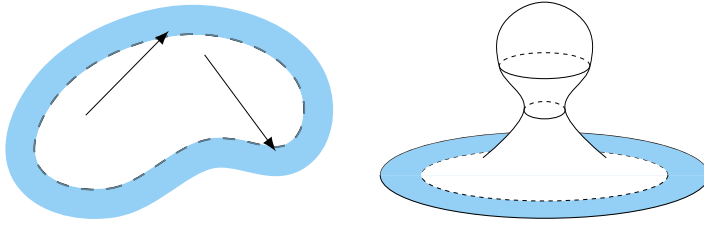


Fig. 1. Figure in the left side $G = I_d$ and the geodesics are straight lines. $a > 0$ on the blue region around the boundary. Figure on the right one has a trapped geodesic that does not meet the damped area in blue which violates the GCC. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

they will meet the region ω . We emphasize that the example presented in the appendix, that satisfies Assumption 1.2, is valid for frictional and Kelvin-Voigt damping.

The examples of situations under which Assumption 1.2 is satisfied presented in the appendix require specific density functions $\rho(x)$ (see Appendix Part II). By the reasons already discussed above, for a general metric $G = (K/\rho)^{-1}$, without extra restrictions on the density, a feedback control region located only in a neighborhood of the boundary of Ω may not be sufficient and uniform decay rate estimates are not expected in general. An optimal way to overcome this difficulty, i.e., minimizing as much as possible the feedback control region, is to follow the ideas presented by Cavalcanti et al. in [21] and [19]. In these articles, the authors show that given a Riemannian manifold (Ω, G) with metric $G = (K/\rho)^{-1}$ and ω a neighborhood of the boundary of Ω , there exists a finite family $\{V_i\}_{i=1, \dots, k}$ of open sets with smooth boundary such that their closures are pairwise disjoint and $V = \bigcup_{i=1}^k V_i$ satisfies

- $meas(V) \geq meas(M \setminus \omega) - \epsilon$ for every $\epsilon > 0$,
- V can be left without damping.

Thus, if we consider a damping acting in ω and in $(\Omega \setminus \omega) \setminus V$, then every geodesic of the metric $G = (K/\rho)^{-1}$ meets the damped area. Hence, there are no geodesics ‘trapped’ within the sets V_i for all $i = 1, \dots, k$ which are free of dissipative effects and Assumption 1.2 is satisfied. Moreover, $(\Omega \setminus \omega) \setminus V$ can be made arbitrarily small, making the volume of the region where the dissipative effect lies as small as possible for a general metric $G = (K/\rho)^{-1}$ (see Fig. 2).

Remark 2. In the next section we apply the Carleman estimates due to Dos Santos Ferreira [29] and Koch and Tataru [37] to show certain situations where the unique continuation property given in the Assumption 1.3 holds. Although the result has been proved for a single equation, a new proof is required with respect to coupled systems of wave equations, which, to our surprise, has not been done so far (at least at the level of the above-mentioned articles). In the present work we are successful in proving a unique continuation property for coupled systems by using sharp $L^p - L^q$ -Carleman estimates. In this direction the authors would like to thank Daniel Tataru for fruitful discussions regarding this issue. However, it is worth mentioning the most recent paper regarding this issue due to Tebou [52]. The author considers a system of two coupled non-conservative wave equations. For this system, he proves several observability estimates. Those observability estimates are sharp in the sense that they lead by duality to the controllability (exact or approximate) of the coupled system with a single control acting through one of the equations

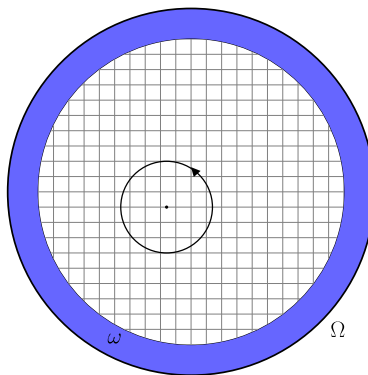


Fig. 2. The region ω (in blue) around the boundary $\partial\Omega$ and $(\Omega \setminus \omega) \setminus V$ (a mesh in light gray) illustrates the damped region on the manifold (Ω, G) , $G = (K/\rho)^{-1}$, which can be considered with measure as small as desired, however totally distributed on Ω . The region $V := \bigcup_{i=1}^k V_k$ (in white) illustrates the region without damping with measure arbitrarily large but also totally distributed in Ω .

only while keeping the same controllability time as for a single equation. Unfortunately the aforementioned result does not seem applicable in the present paper.

In the last section, we consider $(M, \mathbf{g})$ a d -dimensional ($d \geq 2$) compact Riemannian manifold with smooth boundary ∂M where $\mathbf{g}$ denotes a Riemannian metric of class C^∞ . Denoting by $\Delta_{\mathbf{g}}$ the Laplace-Beltrami operator M , we study the asymptotic stability of the following wave system

$$\begin{cases} u_{tt} - \Delta_{\mathbf{g}} u - \gamma(x)v_t + a(x)g(u_t) = 0 & \text{in } M \times (0, +\infty), \\ v_{tt} - \Delta_{\mathbf{g}} v + \gamma(x)u_t = 0 & \text{in } M \times (0, +\infty), \\ u = v = 0 & \text{on } \partial M \times (0, +\infty), \\ u(x, 0) = u_0(x), \quad u_t(x, 0) = u_1(x) & \text{in } M, \\ v(x, 0) = v_0(x), \quad v_t(x, 0) = v_1(x) & \text{in } M, \end{cases} \quad (1.6)$$

where the damping coefficient $a(x)$ is non-negative and effective on an open proper subset M_* (to be defined later) of M . Regarding the coupling coefficient γ we assume that there exist two positive constants c_1 and c_2 such that $a(x)/c_1 \leq \gamma(x) \leq c_2 a(x)$ for a.e. x in M . In particular, the coupling coefficient is effective only where the damping coefficient is effective. Moreover, g is a monotone increasing function such that $A|s| \leq g(s) \leq B|s|$ for all $|s| \geq 1$ and A and B positive constants.

In this case, only one equation in the system is damped and the difficulty lies in determining whether the dissipation effect provided by this single damping is enough to stabilize the whole system, especially since the single damping is nonlinear and effective in a sub-region which, under certain conditions, can be made arbitrarily small. The absence of damping in one of the equations can be compensated by the effect of the strong coupling given by the velocities. However, since the coupling and damping coefficient vanish in the same regions, then in these parts of the domain one of the equation is not only left without damping but in a certain sense is also uncoupled from the damped equation, making it even more difficult to transfer the damping properties from the damped equation to the whole system.

1.2. Main goal, methodology and previous results

The main objective of the present manuscript is to prove the existence and uniqueness for weak solutions to problem (1.1) and, in addition, that those solutions decay exponentially and uniformly to zero, that is, setting

$$E_{u,v}(t) = \frac{1}{2} \int_{\Omega} \rho(x) |u_t(x, t)|^2 + \rho(x) |v_t(x, t)|^2 + \nabla u(x, t)^{\top} \cdot K(x) \cdot \nabla u(x, t) dx \\ + \frac{1}{2} \int_{\Omega} \nabla v(x, t)^{\top} \cdot K(x) \cdot \nabla v(x, t) + (uv)^2(x, t) dx, \quad (1.7)$$

there exist positive constants C, γ , such that

$$E_u(t) \leq C e^{-\gamma t} E_u(0), \text{ for all } t \geq T_0, \quad (1.8)$$

for all weak solutions to problem (1.1) provided that the initial data $\{u_0, v_0, u_1, v_1\}$ are taken in bounded sets of $H_0^1(\Omega) \times H_0^1(\Omega) \times L^2(\Omega) \times L^2(\Omega)$. This result is a *local stabilization result*. Indeed, the constants C and γ are uniform on every ball in $H_0^1(\Omega) \times H_0^1(\Omega) \times L^2(\Omega) \times L^2(\Omega)$ with radius $R > 0$ of the energy space but the result does not guarantee that the decay rate is a global one, i.e. whether (1.8) holds with constants C, γ which are independent of the initial data.

Inspired by Dehman, Gérard, Lebeau, [27] or Dehman, G. Lebeau and Zuazua [28] we give a direct proof of the inverse inequality to problem (1.1), namely, we prove that given $T > 0$ there exists a constant positive $C = C(T)$ such that

$$E_{u,v}(0) \leq C \int_0^T \int_{\Omega} \left(a(x) |u_t(x, t)|^2 + b(x) |v_t(x, t)|^2 \right) dx dt, \quad (1.9)$$

provided the initial data are taken in bounded sets of $H_0^1(\Omega) \times L^2(\Omega)$.

To prove (1.9) and therefore the stability result, we argue by contradiction and we find a sequence of $\{w^n, z^n\}$ of weak solutions to problem (1.1) such that $E_{w^n, z^n}(0) = 1$. In order to obtain a contradiction we need to prove that $E_{w^n, z^n}(0) \rightarrow 0$ as $n \rightarrow +\infty$. We shall prove, by exploiting the properties of $K(x)$, $a(x)$, $b(x)$ and a unique continuation principle, that

$$\int_0^T \int_{\omega} |w_t^n|^2 dx dt \rightarrow 0 \text{ and } \int_0^T \int_{\omega} |z_t^n|^2 dx dt \rightarrow 0 \quad (1.10)$$

when n goes to infinity.

Our wish is to propagate the convergence (1.10) from $\omega \times (0, T)$ to the whole set $\Omega \times (0, T)$. In order to do this, we are able to successfully use microlocal analysis. Indeed, we consider the microlocal defect measure μ , in short m.d.m., firstly introduced by Gérard [34] associated to the solution of the linear wave equation. We note that once one has

$$\begin{aligned}\rho(x)\partial_t^2 w_t^n - \operatorname{div}(K(x)\nabla w_t^n) &\rightarrow 0 \text{ in } H_{loc}^{-1}(\Omega \times (0, T)), \\ \rho(x)\partial_t^2 z_t^n - \operatorname{div}(K(x)\nabla z_t^n) &\rightarrow 0 \text{ in } H_{loc}^{-1}(\Omega \times (0, T)),\end{aligned}\quad (1.11)$$

so, using properties associated to μ we are able to prove that μ propagates along the bicharacteristic flow of the wave operator $\rho(x)\partial_t^2 - \operatorname{div}(K(x)\nabla(\cdot))$ proving the desired convergence. It is important to observe that, in the present article, in order to avoid certain technicalities that may arise when considering the propagation up to the boundary, we consider ω , the region where the damping is effective, as a neighborhood of the boundary and satisfying the GCC. Under these conditions, we can use the propagation results found in [16]. This will be clarified in section 3 and in the appendix of the present manuscript.

Inspired by the works [9], [10], let us come back to our original problem (1.1) having in mind that $\rho \in C^\infty(\Omega)$ and $\alpha_0 \leq \rho(x) \leq \beta_0$. Fix a coordinate system $(x_1, \dots, x_n)$ on (Ω, G) with Riemannian metric $G = \langle \cdot, \cdot \rangle$, denote $G_{ij} = \langle \partial/\partial x_i, \partial/\partial x_j \rangle$, let $G^{ij} = K_{ij}/\rho$ be the inverse matrix of G_{ij} and set

$$\rho = \sqrt{\det(G_{ij})}.$$

The Laplace-Beltrami operator in this coordinate system is given by

$$\Delta_G u = \frac{1}{\sqrt{\det G_{ij}}} \sum_{i,j=1}^n \frac{\partial}{\partial x_i} \left(\sqrt{\det G_{ij}} G^{ij} \frac{\partial u}{\partial x_j} \right) = \frac{1}{\rho(x)} \operatorname{div}(K(x)\nabla u)$$

where ∇ is the usual gradient correspondent to the Euclidean metric on the domain $(x_1, \dots, x_n)$. Consequently,

$$\rho(x)\partial_t^2 u - \operatorname{div}[K(x)\nabla u] = 0 \Leftrightarrow \partial_t^2 u - \Delta_G u = 0, \quad G = (K/\rho)^{-1} \quad (1.12)$$

and the analysis of (1.1) is equivalent to the following derived problem:

$$\begin{cases} u_{tt} - \Delta_G u + \frac{1}{\rho(x)} v^2 u + \frac{1}{\rho(x)} a(x) u_t - \frac{1}{\rho(x)} \gamma(x) v_t = 0 & \text{in } \Omega \times \mathbb{R}_+, \\ v_{tt} - \Delta_G v + \frac{1}{\rho(x)} u^2 v + \frac{1}{\rho(x)} b(x) v_t + \frac{1}{\rho(x)} \gamma(x) u_t = 0 & \text{in } \Omega \times \mathbb{R}_+, \\ u = v = 0 & \text{on } \partial\Omega \times \mathbb{R}_+, \\ u(x, 0) = u_0(x), \quad u_t(x, 0) = u_1(x) & \text{in } \Omega, \\ v(x, 0) = v_0(x), \quad v_t(x, 0) = v_1(x) & \text{in } \Omega. \end{cases}$$

The model proposed in this article is inspired by an equation introduced by Segal in [50], given by

$$\begin{cases} u_{tt} - \Delta u + m_1^2 u + g v^2 u = 0 & \text{in } \Omega \times (0, T), \\ v_{tt} - \Delta v + m_2^2 v + h u^2 v = 0 & \text{in } \Omega \times (0, T), \end{cases} \quad (1.13)$$

which describes the interaction of scalar fields u, v of mass m_1, m_2 , respectively, with interaction constants g and h . This system defines the motion of charged mesons in an electromagnetic field. As the interest of this paper is to make the mathematical analysis, there is no loss of generality if we consider only the case in which $m_1 = m_2 = 0$ and $g = h = 1$.

The problem (1.1), when $m_1 = m_2 = 0$, and $g = h = 1$ with Dirichlet boundary condition on $\partial\Omega$, was studied by Medeiros and Menzala [42]. In this paper, the authors proved the existence and uniqueness of global weak solutions, using Galerkin method, provided that $d \leq 3$. Further generalizations are also given in [43] and [44] by using Galerkin methods, where the authors consider nonlinearities of the form $|v|^{\rho+2}|u|^\rho u$, $|u|^{\rho+2}|v|^\rho v$ and some hypotheses about the coefficient ρ , which are related to the dimension d of the space. Related to global existence and uniqueness of solutions we also have the article [8], where the authors prove the existence of solutions to a Klein-Gordon system with memory and nonlinearities similar to those considered in [43] and [44]. Another generalization for this system can be founded in [25], where the authors consider a $k \times k$ system of Klein-Gordon equations with acoustic boundary conditions.

In [30] Ferreira deduce decay results for the solutions of the system of nonlinear Klein-Gordon equations in $\mathbb{R}^3$, more precisely, if the initial data are taken in $C_0^\infty(\mathbb{R}^3)$ then, every solution of the system decays uniformly in x at the rate $\mathcal{O}(t^{-\frac{3}{2}})$ as $|t| \rightarrow \infty$, and is asymptotic to a free solution at $t = \infty$ or $t = -\infty$. To get these results, they use techniques developed by Morawetz and Strauss.

We also would like to quote the papers [31] and [32]. In these papers, the exponential stability for the system (1.13) is established considering two frictional damping terms $a(x)u_t$ and $b(x)v_t$. However, we emphasize that in [31] and [32] the problem is considered in a homogeneous medium which is easier to be analyzed since the bicharacteristics are straight lines. We also note that in [31] and [32] the author mentions the pioneering work of Ruiz [49], which ensures the unique continuation property for a single equation with potential $V(t, x) \in L^\infty([0, T], L^n(\Omega))$ and the Euclidean metric $G(x) = I_d$, and assumes, without proof, that the result is valid for coupled systems. In [23], Cavalcanti et al. consider a generalized Klein-Gordon system consisting of two coupled semilinear evolution equations of mixed degenerate hyperbolic-parabolic type and boundary conditions are assumed to be dissipative. The authors proved global existence and asymptotic decay rates.

In [33], the authors investigated time decay properties of solutions of the problem (1.13). They show that, in the $L^\infty(\mathbb{R}^3)$ norm, the solutions decay like $\mathcal{O}(t^{-\frac{3}{2}})$ as $t \rightarrow +\infty$ provided the initial data are sufficiently small. After this, the author prove that finite energy solutions of such a system decay in local energy norm as $t \rightarrow +\infty$. To our knowledge, this was the first work to address the existence and uniqueness of solutions to this system using semigroup theory.

The second objective of the article is to show that the solutions of problem (1.6) decay uniformly to zero. To be precise, setting

$$E_{u,v}(t) = \frac{1}{2} \int_M |u_t(x, t)|^2 + |v_t(x, t)|^2 + |\nabla u(x, t)|^2 + |\nabla v(x, t)|^2 dM$$

the energy associated to problem (1.6), we deduce

$$E_{u,v}(t) \leq S\left(\frac{t}{T_0} - 1\right), \quad \forall t > T_0 \quad (1.14)$$

with $\lim_{t \rightarrow 0} S(t) = 0$, where the contraction semigroup $S(t)$ is the solution of the differential equation

$$\frac{d}{dt} S(t) + q(S(t)) = 0, \quad S(0) = E(0) \quad (1.15)$$

and q is a continuous, monotone increasing function.

In order to prove the decay rate estimate given in (1.14), we prove the following inequality

$$E_{u,v}(T) \leq C \int_0^T \int_{\Omega} a(x) \left(|u_t(x,t)|^2 + |g(u_t(x,t))|^2 \right) dx dt, \quad \text{for all } T > T_0, \quad (1.16)$$

where C is a positive constant. The above inequality is the key point to obtain the decay rate estimates given in (1.14) by considering the method developed by Lasiecka and Tataru in [38], from which, it is possible to derive explicit decay rate estimates from the general formula (1.14) (see, for instance, [1], [2], [17] and [20]).

To achieve observability inequality (1.16), we take advantage of two main ingredients. First, we use an intrinsic multiplier constructed by Cavalcanti et al. in [21]. Given $\epsilon > 0$, and u the solution of an initial-value problem with Dirichlet boundary conditions associated to the wave equation

$$u_{tt} - \Delta_{\mathbf{g}} u = f \quad \text{in } M \times (0, +\infty),$$

for any $f \in L^2(0, T; L^2(M))$, the multiplier is given by $\langle \nabla k, \nabla u \rangle$, where $k : M \rightarrow \mathbb{R}$ is a smooth function such that its Hessian $\nabla^2 k$ is closely related to the nonlinear function g on an open subset $V \subset M$ that satisfies $\text{meas}(V) \geq \text{meas}(M) - \epsilon$, $\text{meas}(V \cap \partial M) \geq \text{meas}(\partial M) - \epsilon$ and $\langle \nabla k, \nabla u \rangle < 0$ on $V \cap \partial M$. Applying this special multiplier and following the arguments outlined in [21], we obtain an important energy inequality for each of the wave equations in the system. The second important step is to apply an elliptic multiplier successfully used before by Alabau-Boussouira et al. in [6], in order to benefit from the strong coupling through the velocities to finally obtain observability inequality (1.16).

Systems where only one equation has a damping mechanism have been studied by several authors (see, for instance [3], [4] and [5]), motivated by the fact that it may be less costly than to damp each equation or in some applications it may not be possible to damp each component of the system. A strongly coupled wave-type system as presented in the present article, was studied recently by Alabau-Boussouira et al. [6]. The authors proved, assuming that the coupling coefficient and damping coefficient satisfy the Piecewise Multipliers Geometric Condition, that the whole system decays as fast as the single damped equation. The result established in the present article then generalizes this result to compact manifolds. Observe that the result on the asymptotic stability for problem (1.6) in a Riemannian manifold $(M, \mathbf{g})$ applies in particular considering (Ω, G) with metric $G = (K/\rho)^{-1}$ and by the equivalence relation given in (1.12) to the strongly coupled wave-type system in an inhomogeneous medium. This hyperbolic system was also investigated by Kapitonov in [36] with locally distributed linear damping, under the condition that the support of the coupling and damping coefficient coincide, obtaining the uniform stabilization of the solution.

There are several works concerning the asymptotic behavior of wave equations with frictional damping of the form au_t or $ag(u_t)$ in the Euclidean setting or in the Riemannian one. In [41], dissipative systems with nonlinear localized damping are considered, weakening the usual geometrical condition on the localization of the damping and eliminating the usual assumption on the polynomial growth of the feedback at zero. In Cavalcanti et al. [18], [21], [19], the authors investigated the wave equation with localized nonlinear damping. In the first paper, the authors

consider compact surfaces, while the other two articles deal with compact manifolds. Considering a damping effective in the complement of a set satisfying a technical condition connected to the existence of multipliers, the authors were able to prove uniform and optimal decay rates of the energy in [19], whereas in [21], the optimal decay rate was obtained under arbitrary small control regions based on the construction of a special multiplier. Alabau-Boussouira in [2] proved sharp or quasi-optimal upper energy decay rates for a class of nonlinear feedbacks ranging from very weak non-linear dissipation to polynomial or polynomial-logarithmic decaying feedbacks at the origin. Then in 2011, Alabau-Boussouira and Ammari [6] obtained sharp-energy decay rates for the nonlinearly damped abstract infinite-dimensional systems, combining optimal geometric conditions as provided by Bardos et al. [11] with an optimal-weight convexity method of Alabau-Boussouira (see [2] and [1]), an approach that was later extended to uniform discretization of nonlinearly damped evolution equations in [7]. Several other articles concerning frictional damping are worth mentioning, among them, [14], [24], [26], [39], [46], [47] and [48].

Our paper is organized as follows. In Section 2 we verify that the unique continuation property holds for the coupled system. In Section 3 we give some notation and we establish the well-posedness to problem (1.1). In Section 4 and in the Appendix we give the proof of the stabilization which consists our main result. In Section 5 we prove the stabilization result for problem (1.6).

2. Unique continuation property

Let us now recall some results on the unique continuation property for differential operators. For this, we will use the notations of Dos Santos Ferreira [29] and Koch-Tataru [37].

Let $P(x, D)$ be a second order differential operator of real principal type with C^∞ coefficients and u be the solution of the differential equation

$$P(x, D)u + V(x)u = 0, \quad x \in \Omega \subset \mathbb{R}^d \quad (2.1)$$

where $V \in L_{loc}^{\frac{d}{2}}$ and let S be a C^∞ hypersurface in $\mathbb{R}^d$ locally defined by

$$\begin{aligned} S_+ &= \{x \in \Omega : \psi(x) > \psi(x_0)\} \\ S_- &= \{x \in \Omega : \psi(x) < \psi(x_0)\}. \end{aligned} \quad (2.2)$$

We say that a solution of (2.1) has unique continuation across the hypersurface S if when u vanishes on the side S_- then it vanishes on a whole neighborhood of x_0 .

Definition 2.1. The hypersurface $S = \{x \in \Omega : \phi(x) = \phi(x_0)\}$ is said to be (strictly) pseudoconvex at $x_0 \in \Omega$ with respect to the real principal type differential operator P of order 2 whose principal symbol is p whenever

$$p(x_0, \xi) = H_p \phi(x_0, \xi) = 0 \Rightarrow H_p^2(x_0, \xi) < 0 \text{ for all } \xi \in \mathbb{R}^d.$$

Using $(L^p - L^q)$ Carleman estimates, the following result is proved by Dos Santos Ferreira in [29]

Proposition 2.1. Let $P(x, D)$ be a differential operator of order 2, defined on an open set $\Omega \subset \mathbb{R}^d$, $n > 2$. Suppose that there exists $M \geq 1$, a neighborhood Ω_0 of x_0 and a function $\phi \in C^\infty$ verifying $\{x \in \Omega_0 : x \neq x_0, \phi(x) \leq \phi(x_0)\} \subset S_-$ such that the Carleman estimate

$$\|e^{-\tau\phi} v\|_{L^{\frac{2d}{d-2}}} \leq C \|e^{-\tau\phi} P(x, D)v\|_{L^{\frac{2d}{d+2}}} \quad (2.3)$$

holds for all $u \in C_0^\infty(\Omega_0)$ and $\tau \geq M$. Then if $u \in H^1$ is the solution of the equation (2.1) on Ω where $V \in L^{\frac{d}{2}}_{loc}$ and u vanishes on S_- , then exists a neighborhood of x_0 on which u vanishes.

Inspired in this Proposition, we obtain the following result:

Proposition 2.2. Let $P_1(x, D)$ and $P_2(x, D)$ be two differential operators of order 2, defined on an open set $\Omega \subset \mathbb{R}^d$, $n > 2$. Suppose that there exists $M \geq 1$, a neighborhood Ω_0 of x_0 and a function $\phi \in C^\infty$ verifying $\{x \in \Omega_0 : x \neq x_0, \phi(x) \leq \phi(x_0)\} \subset S_-$ such that the following Carleman estimates

$$\|e^{-\tau\phi} u\|_{L^{\frac{2d}{d-2}}} \leq C \|e^{-\tau\phi} P_1(x, D)u\|_{L^{\frac{2d}{d+2}}} \quad (2.4)$$

and

$$\|e^{-\tau\phi} v\|_{L^{\frac{2d}{d-2}}} \leq C \|e^{-\tau\phi} P_2(x, D)v\|_{L^{\frac{2d}{d+2}}} \quad (2.5)$$

hold for all $u, v \in C_0^\infty(\Omega_0)$ and $\tau \geq M$. Then, the unique continuation property holds for the solution of the coupled system

$$\begin{cases} P_1(x, D)u + V_1u = & V_3v, \\ P_2(x, D)v + V_2v = & V_3u, \end{cases} \quad (2.6)$$

where V_1, V_2 and V_3 are elements of $L^{\frac{d}{2}}_{loc}$.

Proof. Suppose $x_0 = 0$ and $\psi(0) = 0$ and let $\chi \in C_0^\infty(\Omega_0)$ equal 1 on a neighborhood Ω_1 of 0. Applying the Carleman inequality to the functions $w_1 = \chi u$, $w_2 = \chi v$ and observing that $L^{\frac{d}{2}} \cdot L^{\frac{2d}{d-2}} \subset L^{\frac{2d}{d+2}}$, we obtain

$$\begin{aligned} \|e^{-\tau\phi} w_1\|_{L^{\frac{2d}{d-2}}} &\lesssim \|e^{-\tau\phi} P_1(x, D)w_1\|_{L^{\frac{2d}{d+2}}} \\ &\lesssim \|e^{-\tau\phi} P_1(x, D)(\chi u)\|_{L^{\frac{2d}{d+2}}} \\ &\lesssim \|e^{-\tau\phi} (\chi P_1(x, D)u + [P_1, \chi]u)\|_{L^{\frac{2d}{d+2}}} \\ &\lesssim \|V_3 e^{-\tau\phi} \chi v - V_1 e^{-\tau\phi} \chi u + e^{-\tau\phi} [P_1, \chi]u\|_{L^{\frac{2d}{d+2}}} \\ &\lesssim \|V_3\|_{L^{\frac{d}{2}}(K)} \|e^{-\tau\phi} \chi v\|_{L^{\frac{2d}{d-2}}} + \|V_1\|_{L^{\frac{d}{2}}(K)} \|e^{-\tau\phi} \chi u\|_{L^{\frac{2d}{d-2}}} \\ &\quad + \|e^{-\tau\phi} [P_1, \chi]u\|_{L^{\frac{2d}{d+2}}}, \end{aligned}$$

where $K = \text{supp}(\chi)$. That is,

$$\begin{aligned} \|e^{-\tau\phi} w_1\|_{L^{\frac{2d}{d-2}}} &\lesssim \|V_3\|_{L^{\frac{d}{2}}(K)} \|e^{-\tau\phi} \chi v\|_{L^{\frac{2d}{d-2}}} + \|V_1\|_{L^{\frac{d}{2}}(K)} \|e^{-\tau\phi} \chi u\|_{L^{\frac{2d}{d-2}}} \\ &\quad + \|e^{-\tau\phi} [P_1, \chi] u\|_{L^{\frac{2d}{d+2}}}. \end{aligned} \quad (2.7)$$

Analogously,

$$\begin{aligned} \|e^{-\tau\phi} w_2\|_{L^{\frac{2d}{d-2}}} &\lesssim \|V_3\|_{L^{\frac{d}{2}}(K)} \|e^{-\tau\phi} \chi u\|_{L^{\frac{2d}{d-2}}} + \|V_2\|_{L^{\frac{d}{2}}(K)} \|e^{-\tau\phi} \chi v\|_{L^{\frac{2d}{d-2}}} \\ &\quad + \|e^{-\tau\phi} [P_2, \chi] v\|_{L^{\frac{2d}{d+2}}}. \end{aligned} \quad (2.8)$$

Adding inequalities (2.7) and (2.8) and organizing the terms, we obtain

$$\begin{aligned} \|e^{-\tau\phi} \chi u\|_{L^{\frac{2d}{d-2}}} + \|e^{-\tau\phi} \chi v\|_{L^{\frac{2d}{d-2}}} &\lesssim (\|V_1\|_{L^{\frac{d}{2}}(K)} + \|V_3\|_{L^{\frac{d}{2}}(K)}) \|e^{-\tau\phi} \chi u\|_{L^{\frac{2d}{d-2}}} \\ &\quad + (\|V_2\|_{L^{\frac{d}{2}}(K)} + \|V_3\|_{L^{\frac{d}{2}}(K)}) \|e^{-\tau\phi} \chi v\|_{L^{\frac{2d}{d-2}}} \\ &\quad + \|e^{-\tau\phi} [P_1, \chi] u\|_{L^{\frac{2d}{d+2}}} + \|e^{-\tau\phi} [P_2, \chi] v\|_{L^{\frac{2d}{d+2}}}. \end{aligned} \quad (2.9)$$

Then we choose χ with sufficiently small support so that the two first terms on the right hand side of (2.9) may be absorbed in the left hand side term, thus

$$\|e^{-\tau\phi} \chi u\|_{L^{\frac{2d}{d-2}}} + \|e^{-\tau\phi} \chi v\|_{L^{\frac{2d}{d-2}}} \lesssim \|e^{-\tau\phi} [P_1, \chi] u\|_{L^{\frac{2d}{d+2}}} + \|e^{-\tau\phi} [P_2, \chi] v\|_{L^{\frac{2d}{d+2}}}. \quad (2.10)$$

But $[P_1(x, D), \chi]u$ and $[P_2(x, D), \chi]v$ are classical differential operators of order 1 supported in $\text{supp } u \cap \overline{\Omega_0 \setminus \Omega_1} \subset \{\phi > 0\}$ and $\text{supp } v \cap \overline{\Omega_0 \setminus \Omega_1} \subset \{\phi > 0\}$ where one has $\phi > c > 0$, which implies that

$$\begin{aligned} \|e^{-\tau\phi} \chi u\|_{L^{\frac{2d}{d-2}}} + \|e^{-\tau\phi} \chi v\|_{L^{\frac{2d}{d-2}}} &\lesssim e^{-\tau c} (\| [P_1(x, D), \chi] u \|_{L^{\frac{2d}{d+2}}} + \| [P_2(x, D), \chi] v \|_{L^{\frac{2d}{d+2}}}) \\ &\lesssim e^{-\tau c} (\|u\|_{H^1} + \|v\|_{H^1}). \end{aligned}$$

Therefore,

$$\|e^{-\tau(\phi-c)} \chi u\|_{L^{\frac{2d}{d-2}}} + \|e^{-\tau(\phi-c)} \chi v\|_{L^{\frac{2d}{d-2}}} \lesssim \|u\|_{H^1} + \|v\|_{H^1},$$

that is, $\|e^{-\tau(\phi-c)} \chi u\|_{L^{\frac{2d}{d-2}}} + \|e^{-\tau(\phi-c)} \chi v\|_{L^{\frac{2d}{d-2}}}$ is bounded, which is impossible unless $u = v = 0$ when $\phi < c$. This completes the proof of the unique continuation. $\square$

Similar arguments can be applied to prove unique continuation property for second order hyperbolic coupled systems with nonsmooth coefficients and potentials in $L^{\frac{d+1}{2}}$. Indeed, in Koch-Tataru we can find the following result:

Theorem 1. Let P be a second-order hyperbolic operator with C^2 coefficients. Let ϕ be a strictly pseudo-convex function with respect to P . Then for compactly supported u we have

$$\|e^{\tau\phi}u\|_{L^{\frac{2(d+1)}{d-1}} \cap \tau^{-\frac{1}{4}}H_{\tau}^{\frac{1}{2}}} \lesssim \|e^{\tau\phi}P(x, D)u\|_{L^{\frac{2(d+1)}{d+3} + \tau^{\frac{1}{4}}H_{\tau}^{-\frac{1}{4}}}}, \quad \tau > \tau_0. \quad (2.11)$$

Let $X = L^{\frac{2(d+1)}{d-1}}$ and $Y = \tau^{-\frac{1}{4}}H_{\tau}^{\frac{1}{2}}$, then the inequality (2.11) can be rewritten as

$$\|e^{\tau\phi}u\|_{X \cap Y} \lesssim \|e^{\tau\phi}P(x, D)u\|_{X^* + Y^*}, \quad \tau > \tau_0. \quad (2.12)$$

Note that $X \cap Y \hookrightarrow X$, which implies that $X^* \hookrightarrow X^* + Y^*$. From the inequality (2.12) we deduce that,

$$\|e^{\tau\phi}u\|_X \lesssim \|e^{\tau\phi}P(x, D)u\|_{X^*}, \quad \tau > \tau_0. \quad (2.13)$$

Thus, we deduce the following result:

Proposition 2.3. Let $P_1(x, D)$ and $P_2(x, D)$ be two second-order hyperbolic operators with C^2 coefficients. Let ϕ be a strictly pseudo-convex function with respect to P_1 and P_2 . Then the unique continuation property holds for the solution of the coupled system

$$\begin{cases} P_1(x, D)u + V_1u = V_3v \\ P_2(x, D)v + V_2v = V_3u, \end{cases} \quad (2.14)$$

where V_1 , V_2 and V_3 are elements of $L^{\frac{d+1}{2}}$.

3. Well-posedness

We consider the weak phase space

$$\mathcal{H} = H_0^1(\Omega) \times H_0^1(\Omega) \times L^2(\Omega) \times L^2(\Omega),$$

which is endowed with the inner product

$$\begin{aligned} & \langle (u_1, u_2, u_3, u_4), (v_1, v_2, v_3, v_4) \rangle_{\mathcal{H}} \\ &= \int_{\Omega} \nabla u_1^{\top} \cdot K(x) \cdot \nabla v_1 + \nabla u_2^{\top} \cdot K(x) \cdot \nabla v_2 + \rho u_3 v_3 + \rho u_4 v_4 \, dx. \end{aligned}$$

Denoting $W(t) = (u, v, u_t, v_t)$ we may rewrite problem (1.1) as the following Cauchy problem in $\mathcal{H}$

$$\begin{cases} \frac{\partial W}{\partial t}(t) = \mathcal{A}W(t) + \mathcal{F}(W(t)) \\ W(0) = (u_0, v_0, u_1, v_1), \end{cases} \quad (3.1)$$

where the linear unbounded operator $\mathcal{A} : D(\mathcal{A}) \rightarrow \mathcal{H}$ is given by

$$\mathcal{A} = \begin{pmatrix} 0 & 0 & I & 0 \\ 0 & 0 & 0 & I \\ \frac{1}{\rho} \operatorname{div}(K(x) \nabla(\cdot)) & 0 & -\frac{1}{\rho} a(x) I(\cdot) & +\frac{1}{\rho} \gamma(x) I(\cdot) \\ 0 & \frac{1}{\rho} \operatorname{div}(K(x) \nabla(\cdot)) & -\gamma(x) I(\cdot) & -\frac{1}{\rho} b(x) I(\cdot) \end{pmatrix}, \quad (3.2)$$

that is,

$$\begin{aligned} \mathcal{A}(u, v, w, z) \\ = \left(w, z, \frac{1}{\rho} \operatorname{div}(K(x) \nabla u) - \frac{1}{\rho} a(x) w + \frac{1}{\rho} \gamma(x) z, \frac{1}{\rho} \operatorname{div}(K(x) \nabla v) - \frac{1}{\rho} b(x) z - \frac{1}{\rho} \gamma(x) w \right), \end{aligned} \quad (3.3)$$

with domain

$$D(\mathcal{A}) = \left(H^2(\Omega) \cap H_0^1(\Omega) \right)^2 \times \left(H_0^1(\Omega) \right)^2 \quad (3.4)$$

and $\mathcal{F} : \mathcal{H} \rightarrow \mathcal{H}$ is the nonlinear operator

$$\mathcal{F}(u, v, w, z) = \left(0, 0, -\frac{1}{\rho} v^2 u, -\frac{1}{\rho} u^2 v \right). \quad (3.5)$$

Now, we are in conditions to state the well-posedness result for problem (3.1), which ensures that problem (1.1) is globally well-posed.

Theorem 2 (*Global well-posedness*). Assume that the hypotheses on ρ and K are fulfilled and the initial data $(u_0, v_0, u_1, v_1) \in \mathcal{H}$. Then problem (3.1) possesses a unique mild solution $W \in C([0, T]; \mathcal{H})$. Moreover, if $(u_0, v_0, u_1, v_1) \in D(\mathcal{A})$, then the solution is regular.

Proof. First of all the operator $\mathcal{A} : D(\mathcal{A}) \subset \mathcal{H} \rightarrow \mathcal{H}$ defined by (3.4) and (3.5) generates a C_0 -semigroup of contractions e^{At} on the energy space $\mathcal{H}$. Indeed, it is easy to see that for all $(u, v, w, z) \in D(\mathcal{A})$, we have

$$(\mathcal{A}U, U) = - \int_{\Omega} \left(a(x)|w|^2 + b(x)|z|^2 \right) dx \leq 0, \quad (3.6)$$

which show that the operator $\mathcal{A}$ is dissipative. Next, we shall prove that $I - \mathcal{A} = \mathcal{H}$, proving the wellposedness for the linear problem associated by using the well-known Lumer-Phillips theorem. Indeed, for any given $(f, g, h, k) \in \mathcal{H}$, we solve the equation $\mathcal{A}(u, v, w, z) = (f, g, h, k)$ which is recast on the following way

$$\begin{cases} u - w = f \in H_0^1(\Omega), \\ v - z = g \in H_0^1(\Omega), \\ w - \frac{1}{\rho} \operatorname{div}(K(x) \nabla u) + \frac{1}{\rho} a(x) w - \frac{1}{\rho} \gamma(x) z = h \in L^2(\Omega), \\ z - \frac{1}{\rho} \operatorname{div}(K(x) \nabla v) + \frac{1}{\rho} b(x) z + \frac{1}{\rho} \gamma(x) w = k \in L^2(\Omega). \end{cases} \quad (3.7)$$

From (3.7) we obtain

$$\begin{cases} w = u - f, \\ z = v - g. \end{cases} \quad (3.8)$$

Substituting the equations of (3.8) in the last two equations of (3.7), we obtain

$$\begin{cases} u - f - \frac{1}{\rho} \operatorname{div}(K(x) \nabla u) + \frac{1}{\rho} a(x)(u - f) - \frac{1}{\rho} \gamma(x)(v - g) = h, \\ v - g - \frac{1}{\rho} \operatorname{div}(K(x) \nabla v) + \frac{1}{\rho} b(x)(v - g) + \frac{1}{\rho} \gamma(x)(u - f) = k. \end{cases} \quad (3.9)$$

Equivalently,

$$\begin{cases} \rho u - \operatorname{div}(K(x) \nabla u) + a(x)u - \gamma(x)v = a(x)f + \gamma(x)g + \rho(f + h), \\ \rho v - \operatorname{div}(K(x) \nabla v) + b(x)v + \gamma(x)u = b(x)g + \gamma(x)f + b(x)q + \rho(g + k). \end{cases} \quad (3.10)$$

Define

$$a : H_0^1(\Omega) \times H_0^1(\Omega) \times H_0^1(\Omega) \times H_0^1(\Omega) \longrightarrow \mathbb{R}$$

given by

$$\begin{aligned} a((u_1, v_1), (u_2, v_2)) &= \int_{\Omega} \rho u_1 u_2 + (\nabla u_1)^{\top} K(x) \nabla u_2 dx + \int_{\Omega} \rho a u_1 u_2 dx - \int_{\Omega} \rho \gamma v_1 u_2 dx \\ &\quad + \int_{\Omega} \rho v_1 v_2 + (\nabla v_1)^{\top} K(x) \nabla v_2 dx + \int_{\Omega} \rho b u_2 v_2 dx + \int_{\Omega} \rho \gamma u_1 v_2 dx. \end{aligned} \quad (3.11)$$

Then, $a(\cdot, \cdot)$ is bilinear, continuous and coercive. Thanks to the Lax-Milgram Theorem, given

$$(a_0, b_0) = (a(x)f + \gamma(x)g + \rho(f + h), b(x)g + \gamma(x)f + b(x)q + \rho(g + k)),$$

there exists a unique $(u, v) \in H_0^1(\Omega) \times H_0^1(\Omega)$, such that

$$a((u, v), (\chi, \eta)) = \langle a_0, \chi \rangle_{H^{-1}(\Omega), H_0^1(\Omega)} + \langle b_0, \eta \rangle_{H^{-1}(\Omega), H_0^1(\Omega)} \text{ for all } (\chi, \eta) \in H_0^1(\Omega) \times H_0^1(\Omega).$$

Taking $\chi \in C_0^\infty(\Omega)$ and $\eta = 0$ we deduce that

$$\rho u - \operatorname{div}(K(x) \nabla u) = a_0 - au + \gamma v \text{ in } \mathcal{D}'(\Omega)$$

and consequently,

$$\rho u - \operatorname{div}(K(x) \nabla u) = a_0 - au + \gamma v \text{ in } L^2(\Omega).$$

Proceeding in the same way, we deduce that

$$\rho v - \operatorname{div}(K(x)\nabla v) = b_0 - bv - \gamma u \text{ in } L^2(\Omega).$$

From the above we deduce that $(u, v) \in (H^2(\Omega))^2$, and, consequently, $(u, v, w, z) \in D(\mathcal{A})$, which gives us the desired solution. By the Theorem 1.4.6 in Pazy's book [45] we obtain $\overline{D(\mathcal{A})} = \mathcal{H}$. Thus, $\mathcal{A}$ generates a C_0 semigroup of contractions on H by Lumer-Phillips Theorem.

Moreover the nonlinear operator $\mathcal{F} : \mathcal{H} \rightarrow \mathcal{H}$ given in (3.5) is a locally Lipschitz continuous operator. Given a bounded set B in $\mathcal{H}$, there exists $R > 0$ such that $\|U\| \leq R$ for all $U \in B$. Given $U = (u_1, u_2, u_3, u_4)$ and $V = (v_1, v_2, v_3, v_4)$ elements of B , we have that

$$\begin{aligned} \|\mathcal{F}U - \mathcal{F}V\|_{\mathcal{H}}^2 &= \left\| \left(0, 0, -\frac{1}{\rho}(u_2^2 u_1 - v_2^2 v_1), -\frac{1}{\rho}(u_1^2 u_2 - v_1^2 v_2) \right) \right\|_{\mathcal{H}}^2 \\ &= \left\| \frac{1}{\rho}(v_2^2 v_1 - u_2^2 u_1) \right\|_{L^2(\Omega)}^2 + \left\| \frac{1}{\rho}(v_1^2 v_2 - u_1^2 u_2) \right\|_{L^2(\Omega)}^2 \\ &= \left\| \frac{1}{\rho}(v_2^2 v_1 - u_2^2 v_1 + u_2^2 v_1 - u_2^2 u_1) \right\|_{L^2(\Omega)}^2 \\ &\quad + \left\| \frac{1}{\rho}(v_1^2 v_2 - u_1^2 v_2 + u_1^2 v_2 - u_1^2 u_2) \right\|_{L^2(\Omega)}^2 \\ &= \left\| \frac{1}{\rho}((v_2^2 - u_2^2)v_1 + u_2^2(v_1 - u_1)) \right\|_{L^2(\Omega)}^2 \\ &\quad + \left\| \frac{1}{\rho}((v_1^2 - u_1^2)v_2 + u_1^2(v_2 - u_2)) \right\|_{L^2(\Omega)}^2 \\ &\lesssim \left\| \frac{1}{\rho}(v_2^2 - u_2^2)v_1 \right\|_{L^2(\Omega)}^2 + \left\| \frac{1}{\rho}(v_1^2 - u_1^2)v_2 \right\|_{L^2(\Omega)}^2 \\ &\quad + \left\| \frac{1}{\rho}u_2^2(v_1 - u_1) \right\|_{L^2(\Omega)}^2 + \left\| \frac{1}{\rho}u_1^2(v_2 - u_2) \right\|_{L^2(\Omega)}^2. \end{aligned} \quad (3.12)$$

Let us show that each of the terms in (3.12) is limited by a positive constant that depends on R multiplied by $\|U - V\|_{\mathcal{H}}$. For this, let $i, j \in \{1, 2\}$ and consider $i \neq j$, then from the Hölder's inequality we obtain

$$\begin{aligned} \left\| \frac{1}{\rho}(v_i^2 - u_i^2)v_j \right\|_{L^2(\Omega)}^2 &= \int_{\Omega} \frac{1}{\rho^2} |v_i - u_i|^2 |v_i + u_i|^2 |v_j|^2 dx \\ &\lesssim \left(\int_{\Omega} |v_i - u_i|^6 dx \right)^{\frac{1}{3}} \left(\int_{\Omega} |v_i + u_i|^3 |v_j|^3 dx \right)^{\frac{2}{3}} \end{aligned}$$

$$\begin{aligned}
 &\lesssim \left(\int_{\Omega} |v_i - u_i|^6 dx \right)^{\frac{1}{3}} \left[\left(\int_{\Omega} |v_j|^6 dx \right)^{\frac{1}{2}} \left(\int_{\Omega} |v_i + u_i|^6 dx \right)^{\frac{1}{2}} \right]^{\frac{2}{3}} \\
 &\lesssim \|v_j\|_{L^6(\Omega)}^2 \|v_i + u_i\|_{L^6(\Omega)}^2 \|v_i - u_i\|_{L^6(\Omega)}^2 \\
 &\lesssim \|v_j\|_{L^6(\Omega)}^2 \left(\|v_i\|_{L^6(\Omega)}^2 + \|u_i\|_{L^6(\Omega)}^2 \right) \|v_i - u_i\|_{L^6(\Omega)}^2 \\
 &\lesssim \|v_j\|_{H_0^1(\Omega)}^2 \left(\|v_i\|_{H_0^1(\Omega)}^2 + \|u_i\|_{H_0^1(\Omega)}^2 \right) \|v_i - u_i\|_{H_0^1(\Omega)}^2 \\
 &\lesssim 2R^3 \|U - V\|_{\mathcal{H}}^2.
 \end{aligned} \tag{3.13}$$

Proceeding in the same way, we obtain

$$\begin{aligned}
 \left\| \frac{1}{\rho} u_j^2 (v_i - u_i) \right\|_{L^2(\Omega)}^2 &= \int_{\Omega} \frac{1}{\rho^2} |u_j|^4 |v_i - u_i|^2 dx \\
 &\lesssim \left(\int_{\Omega} |u_j|^6 dx \right)^{\frac{2}{3}} \left(\int_{\Omega} |v_i - u_i|^6 dx \right)^{\frac{1}{3}} \\
 &\lesssim \|u_j\|_{L^6(\Omega)}^4 \|v_i - u_i\|_{L^6(\Omega)}^2 \\
 &\lesssim \|u_j\|_{H_0^1(\Omega)}^4 \|v_i - u_i\|_{H_0^1(\Omega)}^2 \\
 &\lesssim R^4 \|v_i - u_i\|_{H_0^1(\Omega)}^2 \\
 &\lesssim R^4 \|U - V\|_{\mathcal{H}}^2.
 \end{aligned} \tag{3.14}$$

From (3.12), (3.13) and (3.14) we obtain

$$\|\mathcal{F}U - \mathcal{F}V\|_{\mathcal{H}}^2 \leq C(R) \|U - V\|_{\mathcal{H}}^2,$$

which proves that $\mathcal{F}$ is locally Lipschitz. Hence by Theorem 6.1.4 and 6.1.5 in Pazy's book [45], the Cauchy problem (1.1) has a unique mild solution

$$W(t) = e^{\mathcal{A}t} W(0) + \int_0^t e^{\mathcal{A}(t-s)} \mathcal{F}(W(s)) ds, \text{ for all } t \in [0, T_{\max}).$$

Let us see that $T_{\max} = \infty$. Indeed, given the energy functional defined in (1.7), it follows that

$$\frac{d}{dt} E_{u,v}(t) = - \int_{\Omega} a(x) |u_t(x, t)|^2 dx - \int_{\Omega} b(x) |v_t(x, t)|^2 dx,$$

which shows that $E_{u,v}(t)$ is non-increasing with $E_{u,v}(t) \leq E_u(0)$ for all $t \in [0, T_{\max})$. On the other hand, we have that

$$\begin{aligned}
E_{u,v}(t) &\geq \frac{1}{2} \int_{\Omega} \rho(x) |u_t(x, t)|^2 + \rho(x) |v_t(x, t)|^2 dx \\
&\quad + \frac{1}{2} \int_{\Omega} \nabla u(x, t)^{\top} \cdot K(x) \cdot \nabla u(x, t) + \nabla v(x, t)^{\top} \cdot K(x) \cdot \nabla v(x, t) dx \\
&= \frac{1}{2} \|(u, v, u_t, v_t)\|_{\mathcal{H}}^2.
\end{aligned}$$

Thus,

$$\frac{1}{2} \|(u, v, u_t, v_t)\|_{\mathcal{H}}^2 \leq E_u(t) \leq E_u(0), \text{ for all } t \in [0, T_{\max}).$$

Therefore the local solutions cannot blow-up in finite time and it follows that $T_{\max} = \infty$. $\square$

4. Exponential stability

In this section we give the proofs of the main result of the present article which reads as follows

Theorem 3. *Under Assumptions presented above, given $R > 0$, there exist constants C and γ such that the following inequality holds*

$$E_{u,v}(t) \leq C e^{-\gamma t} E_{u,v}(0), \quad t > 0, \quad (4.1)$$

for all weak solution to problem (1.1) provided that $E_{u,v}(0) \leq R$.

Remark 3. By standard density arguments it is enough to work with regular solutions at all times since the decay rate estimate given in (4.1) can be recovered for weak solutions as well.

In order to prove Theorem 3 and having in mind that problem (1.1) satisfies the semigroup property, so, in view of the identity of the energy associated to problem (1.1), namely,

$$\begin{aligned}
E_{u,v}(t_2) - E_{u,v}(t_1) &= - \int_{t_1}^{t_2} \int_{\Omega} a(x) |u_t(x, t)|^2 dx dt \\
&\quad - \int_{t_1}^{t_2} \int_{\Omega} b(x) |v_t(x, t)|^2 dx dt, \text{ for all } 0 \leq t_1 \leq t_2,
\end{aligned} \quad (4.2)$$

it is enough to prove that the following observability estimate holds:

Lemma 4.1. *For every $T > 0$ and every $R > 0$, there exists a constant $C > 0$ such that inequality*

$$E_{u,v}(0) \leq C \int_0^T \int_{\Omega} a(x) |u_t(x, t)|^2 + b(x) |v_t(x, t)|^2 dx dt, \quad (4.3)$$

holds for every regular solution u of the damped problem (1.1) if the initial data satisfies

$$E_u(0) \leq R. \quad (4.4)$$

Proof. We argue by contradiction. Let us assume that (4.3) does not hold. Then, there exist a sequence $\{u^n, v^n\}$ of regular solutions to problem, according to Theorem 2, such that the initial data satisfy

$$E_{u^n, v^n}(0) \leq L, \text{ for all } n \in \mathbb{N}, \text{ and for some } L > 0, \quad (4.5)$$

and, in addition,

$$\lim_{n \rightarrow \infty} \frac{E_{u^n, v^n}(0)}{\int_0^T \int_{\Omega} a(x) |u_t^n(x, t)|^2 + b(x) |v_t^n(x, t)|^2 dx dt} = +\infty. \quad (4.6)$$

From (4.6) yields

$$\lim_{n \rightarrow \infty} \frac{\int_0^T \int_{\Omega} a(x) |u_t^n(x, t)|^2 + b(x) |v_t^n(x, t)|^2 dx dt}{E_{u^n, v^n}(0)} = 0. \quad (4.7)$$

Taking (4.5) and (4.7) into account, we deduce

$$\lim_{n \rightarrow \infty} \int_0^T \int_{\Omega} a(x) |u_t^n(x, t)|^2 + b(x) |v_t^n(x, t)|^2 dx dt = 0, \quad (4.8)$$

and since $a(x) \geq a_0 > 0$ a.e. in ω from (4.8) we infer

$$\lim_{n \rightarrow \infty} \int_0^T \int_{\omega} |u_t^n(x, t)|^2 dx dt = 0, \quad (4.9)$$

and once $b(x) \geq b_0 > 0$ a.e. in Ω we also obtain that

$$\lim_{n \rightarrow \infty} \int_0^T \int_{\omega} |v_t^n(x, t)|^2 dx dt = 0. \quad (4.10)$$

Since $E_{u^n, v^n}(t)$ is non-increasing and $E_{u^n, v^n}(0)$ remains bounded then, we obtain a subsequence, still denoted by $\{(u^n, v^n)\}$ which verifies

$$(u^n, v^n) \overset{*}{\rightharpoonup} (u, v) \text{ in } (L^\infty(0, T; H_0^1(\Omega)))^2, \quad (4.11)$$

$$(u_t^n, v_t^n) \overset{*}{\rightharpoonup} (u_t, v_t) \text{ in } (L^\infty(0, T; L^2(\Omega)))^2. \quad (4.12)$$

From standard compactness arguments, see [40] or [51], we deduce, for an eventual subsequence, that will be denoted by the same notation, that

$$(u^n, v^n) \rightarrow (u, v) \text{ in } (L^\infty(0, T; L^q(\Omega)))^2, \text{ for all } q \in [2, 6) \quad (4.13)$$

From (4.13) we infer

$$(v^n)^2 u^n \rightarrow v^2 u \text{ a.e. in } \Omega \times (0, T)$$

and

$$(u^n)^2 v^n \rightarrow u^2 v \text{ a.e. in } \Omega \times (0, T).$$

Moreover, $((v^n)^2 u^n)$ and $((u^n)^2 v^n)$ are bounded in $L^2(0, T; L^2(\Omega))$. Indeed,

$$\begin{aligned} \int_0^T \int_\Omega |(v^n)^2 u^n|^2 dx dt &= \int_0^T \int_\Omega |v^n|^4 |u^n|^2 dx dt \\ &\leq \int_0^T \left(\int_\Omega |v^n|^6 dx \right)^{\frac{2}{3}} \left(\int_\Omega |u^n|^2 dx \right)^{\frac{1}{3}} dt \\ &= \int_0^T \|v^n\|_{L^6(\Omega)}^4 \|u^n\|_{L^6(\Omega)}^2 dt \\ &\leq \|v^n\|_{L^8(0, T; L^6(\Omega))}^4 \|v^n\|_{L^4(0, T; L^6(\Omega))}^2 \\ &\lesssim \|v^n\|_{L^\infty(0, T; H_0^1(\Omega))}^4 \|u^n\|_{L^\infty(0, T; H_0^1(\Omega))}^2 < \infty. \end{aligned}$$

Analogously we obtain the bound for $((u^n)^2 v^n)$. By Lions Lemma, we conclude that

$$(v^n)^2 u^n \rightharpoonup v^2 u \text{ and } (u^n)^2 v^n \rightharpoonup u^2 v \text{ in } L^2(0, T; L^2(\Omega)). \quad (4.14)$$

At this point we shall divide our proof into two cases: (i) $u \neq 0$ and $v \neq 0$ and (ii) $(u, v) = (0, 0)$. We note that these are the only two cases we should consider. Indeed, if $u = 0$ then necessarily $v = 0$ because if we take the following sequence of problems into account

$$\begin{cases} \rho(x) u_{tt}^n - \operatorname{div}(K(x) \nabla u^n) + (v^n)^2 u^n + a(x) u_t^n - \gamma(x) v_t^n = 0 & \text{in } \Omega \times (0, T), \\ \rho(x) v_{tt}^n - \operatorname{div}(K(x) \nabla v^n) + (u^n)^2 v^n + b(x) v_t^n + \gamma(x) u_t^n = 0 & \text{in } \Omega \times (0, T), \\ u = v = 0 & \text{on } \partial\Omega \times (0, T), \\ u^n(x, 0) = u_0^n(x), \quad u_t^n(x, 0) = u_1^n(x) & \text{in } \Omega, \\ v^n(x, 0) = v_0^n(x), \quad v_t^n(x, 0) = v_1^n(x) & \text{in } \Omega, \end{cases} \quad (4.15)$$

and passing to the limit in (4.15) taking (4.7), (4.9), (4.10) and (4.11)–(4.14), into account, having in mind that $0 \leq \gamma(x) \leq a(x)$ and $0 \leq \gamma(x) \leq b(x)$ we obtain, in the distributional sense,

$$\begin{cases} \rho(x) v_{tt} - \operatorname{div}(K(x) \nabla v) = 0 & \text{in } \Omega \times (0, T), \\ v_t = 0 & \text{in } \omega \times (0, T), \end{cases} \quad (4.16)$$

and for $z = v_t$ from (4.16) we infer

$$\begin{cases} \rho(x)z_{tt} - \operatorname{div}(K(x)\nabla z) = 0 & \text{in } \Omega \times (0, T), \\ z = 0 & \text{in } \omega \times (0, T), \end{cases} \quad (4.17)$$

which implies that $z = v_t = 0$ and from (4.16) we deduce that $v = 0$. Analogously if $v = 0$ then $u = 0$.

Case (i): $u \neq 0$ and $v \neq 0$.

Taking the following subsequence of problems into account

$$\begin{cases} \rho(x)u_{tt}^n - \operatorname{div}(K(x)\nabla u^n) + (v^n)^2u^n + a(x)u_t^n - \gamma(x)v_t^n = 0 & \text{in } \Omega \times (0, T), \\ \rho(x)v_{tt}^n - \operatorname{div}(K(x)\nabla v^n) + (u^n)^2v^n + b(x)v_t^n + \gamma(x)u_t^n = 0 & \text{in } \Omega \times (0, T), \\ u^n(x, 0) = u_0^n(x), \quad u_t^n(x, 0) = u_1^n(x) & \text{in } \Omega, \\ v^n(x, 0) = v_0^n(x), \quad v_t^n(x, 0) = v_1^n(x) & \text{in } \Omega, \end{cases} \quad (4.18)$$

and passing to the limit in (4.18) taking (4.7), (4.9), (4.10) and (4.11)–(4.14), into account, since $0 \leq \gamma(x) \leq a(x)$ and $0 \leq \gamma(x) \leq b(x)$, we obtain, in the distributional sense,

$$\begin{cases} \rho(x)u_{tt} - \operatorname{div}(K(x)\nabla u) + v^2u = 0 & \text{in } \Omega \times (0, T), \\ \rho(x)v_{tt} - \operatorname{div}(K(x)\nabla v) + u^2v = 0 & \text{in } \Omega \times (0, T), \\ u_t = v_t = 0 & \text{in } \omega \times (0, T) \end{cases} \quad (4.19)$$

and for $w = u_t$ and $z = v_t$ from (4.19) we infer

$$\begin{cases} \rho(x)w_{tt} - \operatorname{div}(K(x)\nabla w) + 2vuz + v^2w = 0 & \text{in } \Omega \times (0, T), \\ \rho(x)z_{tt} - \operatorname{div}(K(x)\nabla z) + 2uvw + u^2z = 0 & \text{in } \Omega \times (0, T), \\ w = z = 0 & \text{in } \omega \times (0, T). \end{cases} \quad (4.20)$$

Defining $V_1(x, t) = v^2$, $V_2(x, t) = u^2$ and $V_3(x, t) = -2uv$, the system (4.20) can be rewritten as

$$\begin{cases} \rho(x)w_{tt} - \operatorname{div}(K(x)\nabla w) + V_1(x, t)w = V_3(x, t)z & \text{in } \Omega \times (0, T), \\ \rho(x)z_{tt} - \operatorname{div}(K(x)\nabla z) + V_2(x, t)z = V_3(x, t)w & \text{in } \Omega \times (0, T), \\ w = z = 0 & \text{in } \omega \times (0, T). \end{cases} \quad (4.21)$$

Employing Assumption 1.3, we conclude that $(w, z) = (0, 0)$, and, consequently from (4.19) it follows that $u = v = 0$, which is a contradiction.

Case (ii): $(u, v) = (0, 0)$.

Now, we define:

$$\alpha_n := [E_{u^n, v^n}(0)]^{1/2}, \quad w^n := \frac{u^n}{\alpha_n} \quad \text{and} \quad z^n := \frac{v^n}{\alpha_n}. \quad (4.22)$$

Now, let us consider the following subsequence of problems

$$\begin{cases} \rho(x)w_{tt}^n - \operatorname{div}(K(x)\nabla w^n) + \alpha_n^2(z^n)^2 w^n + a(x)w_t^n - \gamma(x)z_t^n = 0 & \text{in } \Omega \times (0, T), \\ \rho(x)z_{tt}^n - \operatorname{div}(K(x)\nabla z^n) + \alpha_n^2(w^n)^2 z^n + b(x)z_t^n + \gamma(x)w_t^n = 0 & \text{in } \Omega \times (0, T), \\ w_n = z_n = 0 & \text{on } \partial\Omega \times (0, T), \\ w^n(x, 0) = w_0^n(x) = \frac{u_0^n}{\alpha_n}, \quad w_t^n(x, 0) = w_1^n(x) = \frac{u_1^n}{\alpha_n} & \text{in } \Omega, \\ z^n(x, 0) = z_0^n(x) = \frac{v_0^n}{\alpha_n}, \quad z_t^n(x, 0) = z_1^n(x) = \frac{v_1^n}{\alpha_n} & \text{in } \Omega. \end{cases} \quad (4.23)$$

A simple calculation shows that

$$E_{w^n, z^n}(t) = \frac{1}{\alpha_n^2} E_{u^n, v^n}(t). \quad (4.24)$$

It is not difficult to check that $E_{w_n, z_n}(0) = 1$ for all $n \in \mathbb{N}$. In order to achieve a contradiction we need to prove that $E_{w^n, z^n}(0)$ converges to zero. Indeed, first, taking (4.8) and (4.22) into consideration we infer

$$\lim_{n \rightarrow \infty} \int_0^T \int_{\Omega} a(x) |w_t^n(x, t)|^2 dx dt = 0, \quad (4.25)$$

and

$$\lim_{n \rightarrow \infty} \int_0^T \int_{\Omega} b(x) |z_t^n(x, t)|^2 dx dt = 0. \quad (4.26)$$

Since $a(x) \geq a_0 > 0$ a.e. in ω from (4.25) we infer

$$\lim_{n \rightarrow \infty} \int_0^T \int_{\omega} |w_t^n(x, t)|^2 dx dt = 0, \quad (4.27)$$

and once $b(x) \geq b_0 > 0$ a.e. in ω from (4.26) we obtain that

$$\lim_{n \rightarrow \infty} \int_0^T \int_{\omega} |z_t^n(x, t)|^2 dx dt = 0. \quad (4.28)$$

Further, from the boundedness $E_{w^n, z^n}(0) \leq C$ we also deduce that for an eventual subsequence of $\{w^n, z^n\}$ that

$$(w^n, z^n) \overset{*}{\rightharpoonup} (w, z) \text{ in } (L^\infty(0, T; H_0^1(\Omega)))^2, \quad (4.29)$$

$$(w_t^n, z_t^n) \overset{*}{\rightharpoonup} (w_t, z_t) \text{ in } (L^\infty(0, T; L^2(\Omega)))^2. \quad (4.30)$$

From standard compactness arguments, we deduce, for an eventual subsequence, that will be denoted by the same notation, that

$$(w^n, z^n) \rightarrow (w, z) \text{ in } (L^\infty(0, T; L^q(\Omega)))^2, \text{ for all } q \in [2, 6) \quad (4.31)$$

Note that, for an eventual subsequence, $\alpha_n \rightarrow \alpha \in [0, +\infty)$. If $\alpha = 0$ and since

$$\alpha_n w^n = u^n \rightarrow 0 \text{ in } L^\infty(0, T; H_0^1(\Omega)), \quad (4.32)$$

$$\alpha_n z^n = v^n \rightarrow 0 \text{ in } L^\infty(0, T; H_0^1(\Omega)), \quad (4.33)$$

taking (4.25), (4.26), (4.27), (4.29), (4.30), (4.32) and (4.33) into account, passing to the limit in (4.23) we arrive at

$$\begin{cases} \rho(x)w_{tt} - \operatorname{div}(K(x)\nabla w) = 0 & \text{in } \Omega \times (0, T), \\ \rho(x)z_{tt} - \operatorname{div}(K(x)\nabla z) = 0 & \text{in } \Omega \times (0, T), \\ w_t = z_t = 0 & \text{in } \omega \times (0, T). \end{cases} \quad (4.34)$$

For $\varphi = w_t$ and $\psi = z_t$ from (4.34) we infer

$$\begin{cases} \rho(x)\varphi_{tt} - \operatorname{div}(K(x)\nabla \varphi) = 0 & \text{in } \Omega \times (0, T), \\ \rho(x)\psi_{tt} - \operatorname{div}(K(x)\nabla \psi) = 0 & \text{in } \Omega \times (0, T), \\ \varphi = \psi = 0 & \text{in } \omega \times (0, T), \end{cases} \quad (4.35)$$

which implies that $\varphi = \psi = 0$ and, consequently from (4.34), $w = z = 0$.

Now, let us consider $\alpha > 0$. So, passing to the limit in (4.23) we arrive at

$$\begin{cases} \rho(x)w_{tt} - \operatorname{div}(K(x)\nabla w) + \alpha^2 z^2 w = 0 & \text{in } \Omega \times (0, T), \\ \rho(x)z_{tt} - \operatorname{div}(K(x)\nabla z) + \alpha^2 w^2 z = 0 & \text{in } \Omega \times (0, T), \\ w_t = z_t = 0 & \text{in } \omega \times (0, T). \end{cases} \quad (4.36)$$

For $\varphi = w_t$ and $\psi = z_t$, from (4.36) we infer

$$\begin{cases} \rho(x)\varphi_{tt} - \operatorname{div}(K(x)\nabla w) + 2\alpha^2 zw\psi + \alpha^2 z^2 \varphi = 0 & \text{in } \Omega \times (0, T), \\ \rho(x)\psi_{tt} - \operatorname{div}(K(x)\nabla \psi) + 2\alpha^2 wz\varphi + \alpha^2 w^2 \psi = 0 & \text{in } \Omega \times (0, T), \\ \varphi = \psi = 0 & \text{in } \omega \times (0, T). \end{cases} \quad (4.37)$$

Denoting $V_1(x, t) = \alpha^2 z^2$, $V_2(x, t) = \alpha^2 w^2$ and $V_3(x, t) = -2\alpha^2 zw$ we can rewrite (4.37) like

$$\begin{cases} \rho(x)\varphi_{tt} - \operatorname{div}(K(x)\nabla w) + V_1(x, t)\varphi = V_3(x, t)\psi & \text{in } \Omega \times (0, T), \\ \rho(x)\psi_{tt} - \operatorname{div}(K(x)\nabla \psi) + V_2(x, t)\psi = V_3(x, t)\varphi & \text{in } \Omega \times (0, T), \\ \varphi = \psi = 0 & \text{in } \omega \times (0, T). \end{cases} \quad (4.38)$$

Employing Assumption 1.3, from (4.38) it follows that $(\varphi, \psi) = (0, 0)$ and, consequently, $w = z = 0$. Thus in any of both cases $w = z = 0$. As a consequence $w = z = 0$ in all the convergences in (4.29)-(4.31).

Proceeding in the same way as in (4.14), and using the fact that $w = z = 0$, we obtain

$$\alpha_n^2(z^n)^2 w^n \rightharpoonup 0 \text{ and } \alpha_n^2(w^n)^2 z^n \rightharpoonup 0 \text{ in } L^2(0, T; L^2(\Omega)). \quad (4.39)$$

We can then use Strichartz estimates (see [13]) to deduce that the sequences $\{z^n\}$ and $\{w^n\}$ are uniformly bounded in $L^4(0, T; L^{12}(\Omega))$. Hence, by applying also Hölder's inequality, we obtain the strong convergence of the sequences $\{\alpha_n^2(z^n)^2 w^n\}$ and $\{\alpha_n^2(w^n)^2 z^n\}$ in $L^2(0, T; L^2(\Omega))$.

Remember that our main objective is to prove that $E_{w^n, z^n}(0)$ converges to zero, where

$$\begin{aligned} E_{w^n, z^n}(t) = & \frac{1}{2} \int_{\Omega} \rho |w_t^n(x, t)|^2 + \rho |z_t^n(x, t)|^2 + (\nabla w^n(x, t))^{\top} \cdot K(x) \cdot \nabla w^n(x, t) dx \\ & + \frac{1}{2} \int_{\Omega} (\nabla z^n(x, t))^{\top} \cdot K(x) \cdot \nabla z^n(x, t) + \alpha_n^2(w^n z^n)^2 dx \end{aligned}$$

Let us use the following notation

$$P := -\rho \partial_t^2 - \sum_{i,j=1}^d \partial_{x_i} (K(x) \partial_{x_j}).$$

Then, we have that given the convergences (4.25), (4.26), (4.29), (4.30) and (4.31) to zero, and having in mind that $0 \leq \gamma(x) \leq a(x)$ and $0 \leq \gamma(x) \leq b(x)$, we deduce that

$$P w_t^n \rightarrow 0 \text{ in } H_{loc}^{-1}(\Omega \times (0, T)) \quad (4.40)$$

and

$$P z_t^n \rightarrow 0 \text{ in } H_{loc}^{-1}(\Omega \times (0, T)). \quad (4.41)$$

Let us also denote by μ_1 and μ_2 the microlocal defect measures associated to $\{w_t^n\}$ and $\{z_t^n\}$ in $L^2(\Omega \times (0, T))$, which is assured by Theorem 5. Then, taking into account Assumption 1.2, we deduce two facts:

(i) The supports of the measures μ_1 and μ_2 are contained in the characteristic set of the wave operator $\{\tau^2 - \frac{1}{\rho} \xi^{\top} \cdot K(x) \cdot \xi = 0\}$.

(ii) μ_1 and μ_2 propagates along the bicharacteristic flow of this operator, which signifies, particularly, that if some point $\omega_0 = (x_0, t_0, \xi_0, \tau_0)$ does not belong to the $\text{supp}(\mu_1)$ or $\text{supp}(\mu_2)$ the whole bicharacteristic issued from ω_0 is out of $\text{supp}(\mu_1)$ or $\text{supp}(\mu_2)$.

Indeed, from (4.40), (4.41) and Theorem 6 we deduce item (i).

Furthermore, from Proposition A.1 and Theorem 8 found in the Appendix, we deduce that $\text{supp}(\mu)$ in $(\Omega \times (0, T)) \times S^d$ is a union of curves like

$$t \in I \cap (0, \infty) \mapsto m \pm(t) = \left(t, x(t), \frac{\pm 1}{\sqrt{1 + |G(x)\dot{x}|^2}}, \frac{\mp G(x)\dot{x}}{\sqrt{1 + |G(x)\dot{x}|^2}} \right), \quad (4.42)$$

where $t \in I \mapsto x(t) \in \Omega$ is a geodesic associated to the metric.

Since $w_t^n \rightarrow 0$ and $z_t^n \rightarrow 0$ in $L^2(\omega \times (0, T))$, we deduce that $\mu_1 = \mu_2 = 0$ in ω and consequently $\text{supp}(\mu_1) \subset (\Omega \setminus \omega) \times (0, T)$ and $\text{supp}(\mu_2) \subset (\Omega \setminus \omega) \times (0, T)$.

On the other hand, let $t_0 \in (0, +\infty)$ and let x be a geodesic of G defined near t_0 . Once the geodesics inside $\Omega \setminus \omega$, enter necessarily in the region ω , then, for any geodesic of the metric G , with $0 \in I$ there exists $t > 0$ such that $m \pm(t)$ does not belong to the $\text{supp}(\mu_1)$ and $\text{supp}(\mu_2)$, so that $m \pm(t_0)$ does not belong as well and item (ii) follows. Once the time t_0 and the geodesic x were taken arbitrary, we conclude that $\text{supp}(\mu_1)$ and $\text{supp}(\mu_2)$ are empty. As a consequence, $w_t^n \rightarrow 0$ and $z_t^n \rightarrow 0$ in $L^2_{loc}(\Omega \times (0, T))$. Since, $v_t^n \rightarrow 0$ and $z_t^n \rightarrow 0$ strongly in $L^2(\omega \times (0, T))$ we deduce

$$w_t^n \rightarrow 0 \text{ in } L^2(0, T; L^2(\Omega)) \quad (4.43)$$

and

$$z_t^n \rightarrow 0 \text{ in } L^2(0, T; L^2(\Omega)). \quad (4.44)$$

It is then easy to show that $E_{w^n, z^n}(0)$ converges to zero. Indeed, let us consider the following cut-off function

$$\theta \in C^\infty(0, T), \quad 0 \leq \theta(t) \leq 1, \quad \theta(t) = 1 \text{ in } (\varepsilon, T - \varepsilon).$$

Multiplying the first equation in (4.23) by $w^n \theta$, the second by $z^n \theta$ and integrating by parts, we infer

$$\begin{aligned} & - \int_0^T \theta(t) \int_{\Omega} \rho(x) |w_t^n|^2 dx dt - \int_0^T \theta'(t) \int_{\Omega} \rho(x) w_t^n w^n dx dt \\ & + \int_0^T \theta(t) \int_{\Omega} (\nabla w^n)^\top \cdot K(x) \cdot \nabla w^n dx dt + \alpha_n^2 \int_0^T \theta(t) \int_{\Omega} (z^n)^2 (w^n)^2 dx dt \\ & + \int_0^T \theta(t) \int_{\Omega} a(x) w_t^n w^n dx dt - \int_0^T \theta(t) \int_{\Omega} \gamma(x) z_t^n w^n dx dt = 0 \end{aligned} \quad (4.45)$$

and

$$\begin{aligned} & - \int_0^T \theta(t) \int_{\Omega} \rho(x) |z_t^n|^2 dx dt - \int_0^T \theta'(t) \int_{\Omega} \rho(x) z_t^n z^n dx dt \\ & + \int_0^T \theta(t) \int_{\Omega} (\nabla z^n)^\top \cdot K(x) \cdot \nabla z^n dx dt + \alpha_n^2 \int_0^T \theta(t) \int_{\Omega} (z^n)^2 (w^n)^2 dx dt \\ & + \int_0^T \theta(t) \int_{\Omega} b(x) z_t^n z^n dx dt + \int_0^T \theta(t) \int_{\Omega} \gamma(x) w_t^n z^n dx dt = 0 \end{aligned} \quad (4.46)$$

Considering the convergences (4.25), (4.26), (4.29)-(4.31) and (4.39) having in mind that $w = z = 0$ from (4.45) and (4.46) we deduce that

$$\lim_{n \rightarrow +\infty} \int_0^T \theta(t) \int_{\Omega} (\nabla w^n)^T \cdot K(x) \cdot \nabla w^n \, dx dt = 0 \quad \text{and}$$

$$\lim_{n \rightarrow +\infty} \int_0^T \theta(t) \int_{\Omega} (\nabla z^n)^T \cdot K(x) \cdot \nabla z^n \, dx dt = 0,$$

which implies that

$$\lim_{n \rightarrow +\infty} \int_{\varepsilon}^{T-\varepsilon} \int_{\Omega} (\nabla w^n)^T \cdot K(x) \cdot \nabla w^n \, dx dt = 0 \quad \text{and}$$

$$\lim_{n \rightarrow +\infty} \int_{\varepsilon}^{T-\varepsilon} \int_{\Omega} (\nabla z^n)^T \cdot K(x) \cdot \nabla z^n \, dx dt = 0.$$

We also have

$$\alpha_n^2 \int_{\varepsilon}^{T-\varepsilon} \int_{\Omega} (w^n z^n)^2 \, dx dt \rightarrow 0.$$

Combining the above convergences we have that

$$\int_{\varepsilon}^{T-\varepsilon} E_{w^n, z^n}(t) \, dt \rightarrow 0.$$

Then by the decrease of the energy, we obtain

$$(T - 2\varepsilon)E_{w^n, z^n}(T - \varepsilon) \rightarrow 0.$$

Combining the energy identity

$$E_{w^n, z^n}(T - \varepsilon) - E_{w^n, z^n}(\varepsilon) = - \int_{\varepsilon}^{T-\varepsilon} a(x)|w_t^n(x, t)|^2 + b(x)|z_t^n(x, t)|^2 \, dx dt,$$

with (4.25), (4.26) and the arbitrariness of $\varepsilon > 0$, it follows that $E_{w^n, z^n}(0) \rightarrow 0$ as $n \rightarrow +\infty$, as we desired to prove. $\square$

Remark 4. Observe that if, instead of assuming that γ satisfies the relation with $a(x)$ and $b(x)$ given in (1.4), we suppose that $\text{supp } \gamma \subset \omega$, the result in Lemma 4.1 still holds and therefore also the stability result in Theorem 3.

5. Uniform stability of the linear model with one nonlinear damping term

Let $(M, \mathbf{g})$ be an compact Riemannian manifold with boundary where $\mathbf{g}$ denotes a Riemannian metric of class C^∞ . In this part, we will investigate the stability of the following system

$$\begin{cases} u_{tt} - \Delta_{\mathbf{g}} u - \gamma(x)v_t + a(x)g(u_t) = 0 & \text{in } M \times (0, +\infty), \\ v_{tt} - \Delta_{\mathbf{g}} v + \gamma(x)u_t = 0 & \text{in } M \times (0, +\infty), \\ u = v = 0 & \text{on } \partial M \times (0, +\infty), \\ u(x, 0) = u_0(x), \quad u_t(x, 0) = u_1(x) & \text{in } M, \\ v(x, 0) = v_0(x), \quad v_t(x, 0) = v_1(x) & \text{in } M. \end{cases} \quad (5.1)$$

We suppose that the feedback function g satisfies the following assumptions:

Assumption 5.1.

- (1) $g(s)$ is continuous and monotone increasing.
- (2) $g(s)s > 0$ for $s \neq 0$,
- (3) $A|s| \leq g(s) \leq B|s|$ for $|s| > 1$,

where A and B are two positive constants.

We assume also the following conditions on the damping and coupling coefficients:

Assumption 5.2. $a, \gamma \in L^\infty(M)$ are nonnegative functions such that

$$a(x) \geq a_0 > 0, \quad \text{almost everywhere on } M_*, \quad (5.2)$$

where M_* is an open set of M that contains $M \setminus V$, and the subset V is referred to Theorem 6.6 in [21]. There exist two positive constants c_1 and c_2 such that

$$a(x) \leq c_1 \gamma(x) \quad \text{and} \quad \gamma(x) \leq c_2 a(x), \quad \forall x \in M.$$

We define the energy of solutions of system (5.1) by

$$E(t) = \underbrace{\frac{1}{2} \int_M (|u_t|^2 + |\nabla u|^2) dM}_{e_1(t)} + \underbrace{\frac{1}{2} \int_M (|v_t|^2 + |\nabla v|^2) dM}_{e_2(t)}.$$

In order to prove a energy decay estimate for (5.1), we start by studying the following system

$$\begin{cases} u_{tt} - \Delta_{\mathbf{g}} u = f & \text{in } M \times (0, +\infty), \\ u = 0 & \text{on } \partial M \times (0, +\infty), \\ u(x, 0) = u_0(x), \quad u_t(x, 0) = u_1(x) & \text{in } M. \end{cases} \quad (5.3)$$

Lemma 5.1. *Under the assumptions above, there exists a positive constant C , such that for all $(u_0, u_1) \in H_0^1(M) \times L^2(M)$, all $f \in L^2((0, \infty); L^2(M))$ and all $T > 0$, the solution of (5.3) satisfies*

$$\int_0^T e(t) dt \leq C e(T) + C \left[\int_0^T \int_M [|f|^2 + a(x)|u_t|^2] dM dt \right], \quad (5.4)$$

$$\text{where } e(t) = \frac{1}{2} \int_M (|u_t|^2 + |\nabla u|^2) dM.$$

Proof. Firstly we recall the result of Theorem 6.6 in [21]: For $\epsilon > 0$, there exist an open set $V \subset M$ and smooth functions $\alpha, k : M \rightarrow \mathbb{R}$ such that $\text{meas}(V) \geq \text{meas}(M) - \epsilon$, $\text{meas}(V \cap \partial M) \geq \text{meas}(\partial M) - \epsilon$, $\nabla \alpha|_V \equiv 0$ and such that α and k satisfy

$$\begin{aligned} C \int_0^T \int_V [u_t^2 + |\nabla u|^2] dM dt &\leq \int_0^T \int_V \left(\frac{\Delta k}{2} - \alpha \right) u_t^2 dM dt \\ &+ \int_0^T \int_V \left[\nabla^2 k(\nabla u, \nabla u) + \left(\alpha - \frac{\Delta k}{2} \right) |\nabla u|^2 \right] dM dt, \end{aligned} \quad (5.5)$$

for some positive constant C and, furthermore,

$$\langle \nabla k, v \rangle < 0 \quad \text{on } \partial M \cap V. \quad (5.6)$$

Using Proposition 4.1, Lemma 4.2 in [21] with $H = \nabla k$, (5.5) and (5.6), we infer

$$\begin{aligned}
 2C \int_0^T e(t) dt &\leq C^* \int_0^T \int_{M/V} [u_t^2 + |\nabla u|^2] dM dt \\
 &+ \left| \left[\int_M u_t \langle \nabla k, \nabla u \rangle dM \right]_0^T \right| + \left| \left[\int_M u_t \alpha u dM \right]_0^T \right| \\
 &+ \left| \int_0^T \int_M f \alpha u dM dt \right| + \left| \int_0^T \int_M f \langle \nabla k, \nabla u \rangle dM dt \right| \\
 &+ \left| \int_0^T \int_{M \setminus V} \langle \nabla u, \nabla \alpha \rangle u dM dt \right| \\
 &+ \frac{1}{2} \left| \int_0^T \int_{\partial M \cap (M \setminus V)} \langle \nabla k, v \rangle |\nabla u|^2 d\partial M dt \right|
 \end{aligned} \tag{5.7}$$

where $C^* = C^*(C, \alpha, k)$.

Denoting

$$\chi = \left[\int_M u_t \langle \nabla k, \nabla u \rangle dM \right]_0^T + \left[\int_M u_t \alpha u dM \right]_0^T. \tag{5.8}$$

By the Cauchy-Schwarz inequality and the Young inequality $ab \leq \frac{a^2}{4v} + vb^2$, where v is a positive number, we obtain

$$\left| \int_0^T \int_M f \langle \nabla k, \nabla u \rangle dM dt \right| \leq \frac{R_1^2}{v} \int_0^T \int_M |f|^2 dM dt + 2v \int_0^T e(t) dt, \tag{5.9}$$

and by the same argument and the Poincaré inequality, we have

$$\left| \int_0^T \int_M f \alpha u dM dt \right| \leq \frac{R_2^2 C_M}{16v} \int_0^T \int_M |f|^2 dM dt + 2v \int_0^T e(t) dt, \tag{5.10}$$

where

$$R_1 = \max_{x \in M} |\nabla f(x)| \quad \text{and} \quad R_2 = \max_{x \in M} |\alpha(x)|.$$

We have

$$\begin{aligned}
\int_0^T \int_{\partial M \cap (M \setminus V)} \langle \nabla k, v \rangle |\nabla u|^2 d\partial M dt &\leq C \int_0^T \int_{M \setminus V} |\nabla u|^2 d\partial M dt \\
&\leq C \int_0^T \int_{\partial M} \langle H_v, v \rangle (\partial_v u)^2 d\partial M dt,
\end{aligned} \tag{5.11}$$

for some positive constant C .

Choosing v sufficiently small and using (5.8)-(5.11) in (5.7) yields

$$\begin{aligned}
\int_0^T e(t) dt &\leq |\chi| + C_1 \int_0^T \int_M |f|^2 dM dt + C_1 \int_0^T \int_{M \setminus V} [a(x)u_t^2 + |\nabla u|^2] dM dt \\
&\quad + C_1 \int_0^T \int_{M \setminus V} |u|^2 dM dt,
\end{aligned} \tag{5.12}$$

where C_1 is a constant that depends on R_1 , R_2 , a_0 , k and Ω .

Let $\tilde{M}$ an extension of M (see Lemma 6.4 in [21] for more details). Notice that we can also extend V and M_* to open subsets $\tilde{V}$ and $\tilde{M}_*$ of $\tilde{M}$, respectively, in a such way that $\tilde{V}$ has smooth boundary and $\tilde{V} \subset \tilde{M}_*$.

Now let $\varepsilon > 0$ such that

$$\tilde{\omega}_\varepsilon := \{x \in \tilde{M}; \text{dist}(x, \partial \tilde{V}) < \varepsilon\} \subset \subset \tilde{M}_*$$

is a tubular neighborhood of $\partial \tilde{V}$ and $\omega_\varepsilon = \tilde{\omega}_\varepsilon \cup M \setminus V$ is contained in M_* .

Following as (7.73)-(7.81) in [21], we obtain the existence of a positive constant $C > 0$ depending on R_1 , R_2 , a_0 , ε and Ω such that

$$\begin{aligned}
\int_0^T e(t) dt &\leq Ce(T) + C \left[\int_0^T \int_M [|f|^2 + a(x)|u_t|^2] dM dt \right] \\
&\quad + C \int_0^T \int_{\omega_\varepsilon} |u|^2 dM dt.
\end{aligned} \tag{5.13}$$

Now, our aim is to absorb the last term on the right-hand side of (5.13). For doing this, we bring idea from [6]. Fix the t variable and consider the solution z of the following elliptic problem in space:

$$\begin{cases} \Delta_{\mathbf{g}} z = \gamma(x)u, & \text{in } M \\ z = 0 & \text{on } \partial M \end{cases} \tag{5.14}$$

Hence, z and z_t satisfy the following estimates

$$\|z\|_{L^2(M)} \leq C \int_M \gamma(x)|u|^2 dM, \quad (5.15)$$

$$\|z_t\|_{L^2(M)} \leq C \int_M \gamma(x)|u_t|^2 dM. \quad (5.16)$$

Multiplying the first equation of (5.3) by z and integrating by parts, we get

$$\begin{aligned} & \int_0^T \int_M \gamma(x)|u|^2 dM dt \\ &= \left[\int_M u_t z dM \right]_0^T + \int_0^T \int_M (-u_t z_t - f z) dM dt. \end{aligned}$$

Hence, using the estimates (5.15)–(5.16) in the above relation, we obtain for all $\varepsilon' > 0$:

$$\begin{aligned} & \int_0^T \int_{\omega_\varepsilon} |u|^2 dM dt \leq \frac{1}{a_0} \int_0^T \int_{M_*} a(x)|u|^2 dM dt \\ & \leq C \int_0^T \int_M \gamma(x)|u|^2 dM dt, \quad (5.17) \\ & \leq C(e(T) + e(0)) + \varepsilon' \int_0^T e(t) dt + \frac{C}{\varepsilon'} \int_0^T \int_M a(x)|u|^2 dM dt \\ & \quad + \frac{C}{\varepsilon'} \int_0^T \int_M |f|^2 dM dt, \\ & = C(2e(T) + \int_0^T \int_M f u_t dM dt) + \varepsilon' \int_0^T e(t) dt + \frac{C}{\varepsilon'} \int_0^T \int_M a(x)|u|^2 dM dt \\ & \quad + \frac{C}{\varepsilon'} \int_0^T \int_M |f|^2 dM dt \\ & \leq 2\varepsilon' \int_0^T e(t) dt + C e(T) + C_{\varepsilon'} \int_0^T \int_M |f|^2 dM dt + C_{\varepsilon'} \int_0^T \int_M a(x)|u_t|^2 dM dt \end{aligned}$$

Inserting (5.17) in (5.13), and choosing ε' sufficiently small, we obtained the desired inequality. $\square$

Proposition 5.1. *For $T > 0$ large enough, the solution of system (5.1) satisfies*

$$E(T) \leq C \int_0^T \int_M \left[a(x) |g(u_t)|^2 + a(x) |u_t|^2 \right] dM dt. \quad (5.18)$$

Proof. Applying Lemma (5.1) to the first equation of (5.1) with $f = \gamma(x)v_t - a(x)g(u_t)$, and $e(t) = e_1(t)$, we infer that

$$\begin{aligned} & \int_0^T e_1(t) dt \\ & \leq C_1 e_1(T) + C_1 \left[\int_0^T \int_M [\gamma^2(x) |v_t|^2 + a^2(x) |g(u_t)|^2 + a(x) |u_t|^2] dM dt \right] \\ & \leq C_1 E(T) + C_1 \left[\int_0^T \int_M [\gamma^2(x) |v_t|^2 + a(x) |g(u_t)|^2 + a(x) |u_t|^2] dM dt \right]. \end{aligned} \quad (5.19)$$

Similarly, we apply Lemma (5.1) to the second equation of (5.1) with $f = -\gamma(x)u_t$, and $e(t) = e_2(t)$, we deduce that

$$\begin{aligned} & \int_0^T e_2(t) dt \leq C_2 e_2(T) + C_2 \left[\int_0^T \int_M [\gamma^2(x) |u_t|^2 + a(x) |v_t|^2] dM dt \right] \\ & \leq C_2 E(T) + C_2 \left[\int_0^T \int_M [a(x) |u_t|^2 + \gamma(x) |v_t|^2] dM dt \right], \end{aligned} \quad (5.20)$$

where we have used, successively, that $\gamma(x) \leq c_2 a(x)$, and $a(x) \leq c_1 \gamma(x)$, for all $x \in M$.

Let $\delta > 0$ be a real parameter that will be chosen later. Adding (5.19) to (5.20), we obtain that for all $T > 0$, and $\delta > 0$

$$\begin{aligned} & \int_0^T (e_1(t) + \delta e_2(t)) dt \leq C(1 + \delta) E(T) + C(\gamma_+ + \delta) \int_0^T \int_M \gamma(x) |v_t|^2 dM dt \\ & \quad + C \int_0^T \int_M a(x) [|g(u_t)|^2 + |u_t|^2] dM dt \end{aligned} \quad (5.21)$$

Next, we estimate the term $\int_0^T \int_M \gamma(x)|v_t|^2 dM dt$ through the coupling relation. Multiplying the first equation in (5.1) by v_t and the second one by u_t and integrating the sum over Ω , we obtain

$$\int_0^T \int_M [v_t(u_{tt} - \Delta_{\mathbf{g}}u - \gamma(x)v_t + a(x)g(u_t)) + u_t(v_{tt} - \Delta_{\mathbf{g}}v + \gamma(x)u_t)] dx dt.$$

After some integrations by parts and using the Cauchy-Schwartz inequality, we find that for all $\epsilon_2 > 0$

$$\begin{aligned} \int_0^T \int_M \gamma(x)|v_t|^2 dM dt &= \left[\int_{\Omega} (u_t v_t + \langle \nabla u, \nabla v \rangle) dx \right]_0^T + \int_0^T \int_M a(x)g(u_t)v_t dx dt \\ &\quad + \int_0^T \int_M \gamma(x)|u_t|^2 dx dt \\ &\leq CE(T) + \epsilon_2 \int_0^T \int_M |v_t|^2 dx dt + \frac{C}{\epsilon_2} \int_0^T \int_M a(x)|g(u_t)|^2 dx dt \\ &\quad + \int_0^T \int_M \gamma(x)|u_t|^2 dx dt. \end{aligned} \quad (5.22)$$

Combining (5.22) and (5.21) and the fact $\gamma(x) \leq c_2 a(x)$, we obtain for all $0 < \delta \leq 1$ and all $\epsilon_2 > 0$ that

$$\begin{aligned} &\int_0^T (e_1(t) + \delta e_2(t)) dt \\ &\leq C(1 + \delta + \gamma_+)E(T) + C \left(1 + (\delta + \gamma_+) \frac{1}{\epsilon_2} \right) \int_0^T \int_M a(x)|g(u_t)|^2 dx dt \\ &\quad + C(\delta + \gamma_+)\epsilon_2 \int_0^T \int_M |v_t|^2 dx dt + C(1 + \delta + \gamma_+) \int_0^T \int_M a(x)|u_t|^2 dx dt. \end{aligned} \quad (5.23)$$

Finally, choosing δ and ϵ_2 so that

$$\delta - C(\delta + \gamma_+)\epsilon_2 > \frac{\delta}{2},$$

we can deduce that

$$\begin{aligned} TE(t) \leq \int_0^T E(t) dt \leq C(1 + \delta + \gamma_+)E(T) + C \left(1 + (\delta + \gamma_+) \frac{1}{\epsilon_2} \right) \int_0^T \int_M a(x) |g(u_t)|^2 dx dt \\ + C(1 + \delta + \gamma_+) \int_0^T \int_M a(x) |u_t|^2 dx dt. \end{aligned} \quad (5.24)$$

Choosing T large enough, we get the desired result. $\square$

To obtain our stability result, we follow the ideas introduced in the literature by Lasiecka and Tataru [38] for attractive forces and expanded by Cavalcanti, Domingos Cavalcanti and Lasiecka [17] for repulsive forces (sources). To this end, we define some needed functions.

Let h be a concave, strictly increasing function, with $h(0) = 0$, satisfying

$$h(s h(s)) \geq s^2 + g^2(s), \text{ for } |s| < 1. \quad (5.25)$$

Note that such function can be straightforwardly constructed, given the hypotheses on function g given in Assumption 5.1. With this function, we define

$$r(\cdot) = h\left(\frac{\cdot}{\text{meas}(Q_T)}\right), \quad Q_T = \Omega \times (0, T), \text{ associated with problem (5.1).} \quad (5.26)$$

As r is monotone increasing, then $cI + r$ is invertible for all $c \geq 0$. For L a positive constant, we then set, respectively,

$$p(x) = (cI + r)^{-1}(Lx), \quad L := (C \text{meas}(Q_T))^{-1}, \quad (5.27)$$

where C is a positive constant that will be determined later.

The function p is easily seen to be positive, continuous and strictly increasing with $p(0) = 0$. Finally, let

$$q(x) = x - (I + p)^{-1}(x). \quad (5.28)$$

Following the arguments presented in [38], we deduce from the observability inequality obtained in the previous proposition, the following stability result for problem (5.1).

Theorem 4. Assume that Assumptions 5.1 and 5.2 are in place and consider the energy $E(t)$ associated to problem (5.1). Then, there exists a $T_0 > 0$ such that

$$E(t) \leq S\left(\frac{t}{T_0} - 1\right), \quad \forall t > T_0, \quad (5.29)$$

with $\lim_{t \rightarrow \infty} S(t) = 0$, where the contraction semigroup $S(t)$ is the solution of the differential equation

$$\frac{d}{dt}S(t) + q(S(t)) = 0, \quad S(0) = E(0), \quad (5.30)$$

where q is as given in (5.28). Here the constant c (from definition (5.27)) is $c \equiv \frac{k_1^{-1} + k_2}{\text{meas}(Q_T)(1 + \|a\|_\infty)}$.

We mention that, by exploiting the inequality (5.24) and based on section 8 in [22], it is possible to derive explicit decay rate estimates (see examples 1, 2, 3 and 4 given in the mentioned section for more details). We mention also that similar estimates as well as different ones can be also obtained from the method firstly developed in [6] and references therein.

Appendix A

A.1. Part I

We start this section by presenting some classical tools regarding the Hamiltonian vector field and its bicharacteristic curves in the cotangent (x, ξ) –space for real functions $p(x, \xi)$.

Let $p \in C^\infty(\Omega \times \mathbb{R}^d \setminus \{0\})$ be a real valued function. We call H_p an Hamiltonian field of p , the following vector field defined in $\Omega \times \mathbb{R}^d \setminus \{0\}$:

$$H_p(x, \xi) = \left(\frac{\partial p}{\partial \xi_1}(x, \xi), \dots, \frac{\partial p}{\partial \xi_n}(x, \xi); -\frac{\partial p}{\partial x_1}(x, \xi), \dots, -\frac{\partial p}{\partial x_n}(x, \xi) \right).$$

The Lie derivative of a function f with respect the Hamiltonian field H_p is given by $H_p(f) = \{p, f\}$, where

$$\{p, f\}(x, \xi) = \sum_{j=1}^d \left(\frac{\partial p}{\partial \xi_j} \frac{\partial f}{\partial x_j} - \frac{\partial p}{\partial x_j} \frac{\partial f}{\partial \xi_j} \right).$$

An Hamiltonian curve of p is an integrable curve of the vector field H_p , that is a maximal solution $s \in I \mapsto (x(s), \xi(s))$ to the Hamilton-Jacobi equations

$$\begin{cases} \dot{x} = p_\xi(x, \xi) = \frac{\partial p}{\partial \xi}(x, \xi), & \dot{\xi} = -p_x(x, \xi) = -\frac{\partial p}{\partial x}, \end{cases} \quad (\text{A.1})$$

where I is an open interval of $\mathbb{R}$.

Since $H_p p = 0$ the function p keeps a constant length value on each of its Hamiltonian curves. We say that such curve is a bicharacteristic of p if this value is null.

Remark 5. Let λ be a C^∞ function on $T^0\Omega$ with real values different of zero. Since

$$H_{\lambda p} = \lambda H_p + p H_\lambda = \lambda H_p \quad \text{if } p = 0,$$

it results that the bicharacteristics of λp and p coincide (modulo a re-parametrization).

Next, for the readers comprehension and for the sake of clarify, we shall announce some results due to Burq and Gérard [16] (without their proofs) which were developed firstly in Gérard [34].

We have the first important result:

Theorem 5. Let $\{u_n\}_{n \in \mathbb{N}}$ be a bounded sequence in $L^2_{loc}(\Omega)$ such that it converges weakly to zero in $L^2_{loc}(\Omega)$. Then, there exists a subsequence $\{u_{\varphi(n)}\}$ and a positive Radon measure μ on $T^1\Omega := \Omega \times S^{d-1}$ such that for all pseudo-differential operator A of order 0 on Ω which admits a principal symbol $\sigma_0(A)$ and for all $\chi \in C_0^\infty(\Omega)$ such that $\chi\sigma_0(A) = \sigma_0(A)$, one has

$$(A(\chi u_{\varphi(n)}), \chi u_{\varphi(n)})_{L^2} \xrightarrow{n \rightarrow +\infty} \int_{\Omega \times S^{n-1}} \sigma_0(A)(x, \xi) d\mu(x, \xi). \quad (\text{A.2})$$

Definition A.1. Under the circumstances of Theorem 5 μ is called the **microlocal defect measure** of the sequence $\{u_{\varphi(n)}\}_{n \in \mathbb{N}}$.

Remark 6. Theorem 5 assures that for all bounded sequence $\{u_n\}_{n \in \mathbb{N}}$ of $L^2_{loc}(\Omega)$ which converges weakly to zero, the existence of a subsequence admitting a microlocal defect measure (in short, m.d.m.). We observe that from (A.2) in the particular case when $A = f \in C_0^\infty(\Omega)$, it follows that

$$\int_{\Omega} f(x) |u_{\varphi(n)}(x)|^2 dx \rightarrow \int_{\Omega \times S^{d-1}} f(x) d\mu(x, \xi), \quad (\text{A.3})$$

so that $u_{\varphi(n)}$ converges to 0 strongly if and only if $\mu = 0$.

The second important result reads as follows:

Theorem 6. Let P be a differential operator of order m on Ω and let $\{u_n\}$ a bounded sequence of $L^2_{loc}(\Omega)$ which converges weakly to 0 and admits a m.d.m. μ . The following statement are equivalents:

- (i) $Pu_n \xrightarrow{n \rightarrow +\infty} 0$ strongly in $H^{-m}_{loc}(\Omega)$ ($m > 0$).
- (ii) $\text{supp}(\mu) \subset \{(x, \xi) \in \Omega \times S^{d-1} : \sigma_m(P)(x, \xi) = 0\}$.

Theorem 7. Let P be a differential operator of order m on Ω , verifying $P^* = P$, and let $\{u_n\}$ be a bounded sequence in $L^2_{loc}(\Omega)$ which converges weakly to 0 and it admits a m.d.m. μ . Let us assume that $Pu_n \xrightarrow{n \rightarrow +\infty} 0$ strongly in H^{1-m}_{loc} . Then, for all function $a \in C^\infty(\Omega \times (\mathbb{R}^d) \setminus \{0\})$ homogeneous of degree $1 - m$ in the second variable and with compact support in the first one,

$$\int_{\Omega \times S^{d-1}} \{a, p\}(x, \xi) d\mu(x, \xi) = 0. \quad (\text{A.4})$$

Finally we present the last important result we need, namely:

Theorem 8. Let P be a self-adjoint differential operator of order m on Ω which admits a principal symbol p . Let $\{u_n\}_n$ be a bounded sequence in $L^2_{loc}(\Omega)$ which converges weakly

to zero, with a microlocal defect measurement μ . Let us assume that Pu_n converges to 0 in $H_{loc}^{-(m-1)}$. Then the support of μ , $\text{supp}(\mu)$, is a union of curves like $s \in I \mapsto \left(x(s), \frac{\xi(s)}{|\xi(s)|}\right)$, where $s \in I \mapsto (x(s), \xi(s))$ is a bicharacteristic of p .

Let us consider the wave operator in an inhomogeneous medium:

$$\rho(x)\partial_t^2 - \sum_{j=1}^n \partial_{x_j} [K(x)\partial_{x_j}].$$

Using the notation $D_j = \frac{1}{i}\partial_j$ we can write

$$P(t, x, D_t, D_x) = -\rho(x)D_t^2 + \sum_{j=1}^n D_{x_j} [K(x)D_{x_j}], \quad D_x = (D_{x_1}, \dots, D_{x_n}),$$

whose principal symbol $p(t, x, \tau, \xi)$ is given by

$$p(t, x, \tau, \xi) = -\rho(x)\tau^2 + K(x)\xi \cdot \xi, \quad \xi = (\xi_1, \dots, \xi_n), \quad (\text{A.5})$$

where $t \in \mathbb{R}$, $x \in \Omega \subset \mathbb{R}^d$, $(\tau, \xi) \in \mathbb{R} \times \mathbb{R}^d$.

Next, we shall describe the bicharacteristics of p . These do not change if we multiply p by a non-null function, rather study the Hamiltonian curves of

$$\tilde{p}(t, x, \tau, \xi) = \frac{1}{2} \left(\frac{K(x)}{\rho(x)} \xi \cdot \xi - \tau^2 \right). \quad (\text{A.6})$$

We have

$$\begin{cases} \dot{t} = \frac{\partial \tilde{p}}{\partial \tau} = -\tau, \\ \dot{x} = \frac{\partial \tilde{p}}{\partial \xi} = \frac{K(x)}{\rho(x)} \xi, \\ \dot{\tau} = -\frac{\partial \tilde{p}}{\partial t} = 0, \\ \dot{\xi} = -\frac{\partial \tilde{p}}{\partial x} = -\frac{1}{2} \nabla \left(\frac{K(x)}{\rho(x)} \right) (\xi \cdot \xi). \end{cases} \quad (\text{A.7})$$

Introducing the function $G(x) := \left(\frac{K(x)}{\rho(x)} \right)^{-1}$, equations in (x, ξ) become in

$$\dot{x} = K(x)\xi \iff \xi = G(x)\dot{x}, \quad (\text{A.8})$$

which implies, considering the notation $\langle x, \xi \rangle = x \cdot \xi$, that

$$\dot{\xi} = (G(x)\dot{x})' = \frac{1}{2} \nabla G(x) \dot{x} \cdot \dot{x}.$$

Once $\tilde{p}$ is null on each the bicharacteristic curves from (A.5) we deduce that

$$\frac{K(x)}{\rho(x)} \xi \cdot \xi = \tau^2 = \text{constant on the curve.} \quad (\text{A.9})$$

On the other hand by (A.8) and (A.9), one has

$$G(x) \dot{x} \cdot \dot{x} = \frac{K(x)}{\rho(x)} \xi \cdot \xi. \quad (\text{A.10})$$

Combining (A.9) and (A.10) we deduce

$$G(x) \dot{x} \cdot \dot{x} = \frac{K(x)}{\rho(x)} \xi \cdot \xi = \tau^2 = \text{constant on the curve,} \quad (\text{A.11})$$

i.e., the quantity $G(x) \dot{x} \cdot \dot{x}$ is preserved under the flow.

By (A.9) and (A.11) we can write

$$\frac{d}{ds} \frac{G(x) \dot{x}}{\sqrt{G(x) \dot{x} \cdot \dot{x}}} = \frac{1}{2} \frac{\nabla G(x) \dot{x} \cdot \dot{x}}{\sqrt{G(x) \dot{x} \cdot \dot{x}}}. \quad (\text{A.12})$$

Setting

$$L(x, \dot{x}) := \sqrt{G(x) \dot{x} \cdot \dot{x}},$$

from the last identity yields

$$\frac{d}{ds} \frac{\partial}{\partial \dot{x}} L(x, \dot{x}) = \frac{\partial}{\partial x} L(x, \dot{x}),$$

which is the Euler-Lagrange equation associated to L , namely, the geodesic equation for the metric G of Ω .

Reversely, if

$$\alpha \mapsto x(\alpha),$$

is a geodesic for the metric G on Ω , and parameterizing the curve x by the curvilinear abscissa σ defined by

$$\frac{d\sigma}{d\alpha} = \sqrt{G(x(\alpha)) \dot{x}(\alpha) \cdot \dot{x}(\alpha)}, \quad (\text{A.13})$$

the equation (A.12) becomes

$$\frac{d}{d\tau} \left(G(x) \frac{dx}{d\tau} \right) = \frac{1}{2} \nabla G(x) \frac{dx}{d\tau} \cdot \frac{dx}{d\tau}, \quad (\text{A.14})$$

with

$$G(x) \frac{dx}{d\sigma} \cdot \frac{dx}{d\sigma} = 1.$$

We find so (A.7), (A.8) and (A.9) setting, for instance, $s = -\frac{\sigma}{\tau}$. We observe that $\frac{dt}{d\sigma} = 1$. As a conclusion, we have proved the following result:

Proposition A.1. *Unless a change of variables, the bicharacteristics of (A.5) are curves of the form*

$$t \mapsto \left(t, x(t), \tau, -\tau \left(\frac{K(x(t))}{\rho(x(t))} \right)^{-1} \dot{x}(t) \right),$$

where $t \mapsto x(t)$ is a geodesic of the metric $G = (A)^{-1} = (g_{ij})$ on Ω , parameterized by the curvilinear abscissa.

A.2. Part II

In what follows we shall give examples of metrics $G = \left(\frac{K}{\rho}\right)^{-1}$ such that its bicharacteristics enter in the region ω where $a(x) > 0$. We consider a Riemannian manifold M with Riemannian metric $G = \langle \cdot, \cdot \rangle$ and Riemannian connection $\tilde{\nabla}$. We denote its Laplace-Beltrami operator by $\tilde{\Delta}$. Fix a coordinate system $(x_1, \dots, x_n)$ on M , denote $G_{ij} = \langle \partial/\partial x_i, \partial/\partial x_j \rangle$, let $G^{ij} = K_{ij}/\rho$ be the inverse matrix of G_{ij} and set $\rho = \sqrt{\det(G_{ij})}$. The Laplace-Beltrami operator in this coordinate system is given by

$$\tilde{\Delta}u = \frac{1}{\sqrt{\det G_{ij}}} \sum_{i,j=1}^n \frac{\partial}{\partial x_i} \left(\sqrt{\det G_{ij}} G^{ij} \frac{\partial u}{\partial x_j} \right) = \frac{1}{\rho(x)} \operatorname{div}(K(x) \nabla u)$$

where ∇ is the usual gradient correspondent to the Euclidean metric on the domain $(x_1, \dots, x_n)$.

Remark 7. It is straightforward that $\det K = \rho^{n-2}$. For $n \geq 3$, it implies that there exist a bijection between the Riemannian metrics G and the operators K given by $G \mapsto K = \rho G^{-1}$ and $K \mapsto G = (\det K)^{\frac{1}{n-2}} K^{-1}$. But for $n = 2$, we always have $\det K = 1$ and K does not determine the Riemannian metric. In fact, observe that we have $K = Id$ in the hyperbolic plane $\mathbb{H}^2$ given in Example A.1 below and in the half plane $\mathbb{R}_+^2$ endowed with Euclidean metric. Although they have the same K , they are not isometric because $\mathbb{H}^2$ have constant Gaussian curvature equal to -1 and the Euclidean open subset $\mathbb{R}_+^2$ have constant Gaussian curvature equal to 0.

The Hessian of a smooth function $\phi : M \rightarrow \mathbb{R}$ is a symmetric 2-form on M defined as

$$\tilde{\nabla}^2 \phi(X, Y) = XY(\phi) - \tilde{\nabla}_X Y(\phi), \quad (\text{A.15})$$

where X and Y are vector fields on M . Here $X(\phi)$ denotes the directional derivative of ϕ in the direction of the vector field X . It is well known that the value of $\nabla^2(X, Y)(p)$ depends only on the values of X and Y on p . It means that the right-hand side of (A.15) does not depend on the smooth extension we take for $X(p)$ and $Y(p)$.

A curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ is a geodesic if $\tilde{\nabla}_{\gamma'(t)}\gamma'(t) \equiv 0$.

The next lemma states the main technical condition we need in order to let some subsets outside ω without damping.

Lemma A.1. *Let M be a complete Riemannian manifold, eventually with boundary, and let $\phi : \text{int}M \rightarrow \mathbb{R}$ be a smooth function. Suppose that ϕ is bounded and $\nabla^2\phi(v, v) \geq c\|v\|^2$ on an open subset $U \subset \text{int}M$, where $c > 0$ is a constant. Then any geodesic on U hits ∂U .*

Proof. Let γ be a geodesic on U . Then

$$\begin{aligned} \frac{d^2}{dt^2}\phi(\gamma(t)) &= \gamma'(t)\gamma'(t)\phi \\ &= \tilde{\nabla}^2\phi(\gamma'(t), \gamma'(t)) + (\tilde{\nabla}_{\gamma'(t)}\gamma'(t))(\phi) \\ &= \tilde{\nabla}^2\phi(\gamma'(t), \gamma'(t)) \geq c\|\gamma'(t)\|^2 \end{aligned}$$

where the last equality holds because γ is a geodesic. Observe that the last term is a positive real constant because $\|\gamma'(t)\|$ does not depend on t .

Then $\phi(\gamma(t))$ is a smooth real valued function which second derivative is bounded below by a strictly positive constant. Then it is not difficult to prove that if γ is defined on a interval (a, ∞) , then $\lim_{t \rightarrow \infty} \phi(\gamma(t)) = \infty$. Analogously $\lim_{t \rightarrow -\infty} \phi(\gamma(t)) = \infty$ whenever γ is defined on a interval $(-\infty, a)$. But neither of the cases are possible if γ remain forever in U because ϕ is bounded there. Therefore γ must hit ∂U . $\square$

We would like to know if $W \subset \Omega - \omega$ can be left without damping. If $\bar{W}$ is contained in an open subset $V \subset \text{int}\Omega$ with Riemannian metric G and $\phi : V \rightarrow \mathbb{R}$ is a smooth function such that $\tilde{\nabla}^2\phi(v, v) > 0$ for every $v \neq 0$, then W can be left without damping. In fact, we are able to find an open subset $U \subset\subset V$ containing W , $\phi|_U$ is bounded, satisfies $\tilde{\nabla}^2(\phi|_U)(v, v) \geq c\|v\|^2$ for some $c > 0$ and we can use Lemma A.1 for $\phi|_U$.

A particular a common instance is the case when $W_1 \subset V_1, \dots, W_m \subset V_m$ is a pairwise disjoint family of compact subsets in $\Omega - \omega$, where each W_i satisfies the conditions of W of the former paragraph and $\phi_i : V_i \rightarrow \mathbb{R}$ are the respective smooth functions with positive Hessian. We can choose $V_1, \dots, V_m$ such that they are pairwise disjoint. Let $U_i \subset\subset V_i$ be open subsets containing W_i . Due to the differentiable version of the Urysohn's lemma, there exist a differentiable function $\eta : \Omega \rightarrow \mathbb{R}$ such that $\eta \equiv 1$ on $\cup_i W_i$ and $\eta \equiv 0$ on $\Omega - \cup_i U_i$. Now define

$$\phi(x) = \begin{cases} \eta(x)\phi_i(x) & \text{if } x \in V_i \\ 0 & \text{otherwise.} \end{cases}$$

Then ϕ is a bounded smooth function which has positive Hessian in a neighborhood of $\cup_i W_i$ and we have proved the following useful result:

Proposition A.2. *Let $\{W_i\}_{i=1, \dots, m}$ a family of pairwise disjoint compact subsets of $V - \omega$. Suppose that there exist neighborhoods V_i of W_i and smooth functions $\phi : V_i \rightarrow \mathbb{R}$ with positive Hessian. Then $\cup_i W_i$ can be left without damping.*

Of course $W = \Omega - \omega$ is a particular case of Proposition A.2.

Let M be a Riemannian manifold and $p \in M$. Let $d : M \rightarrow \mathbb{R}$ be the distance function from p . Then d^2 is a smooth function defined on the ball $B(p, r)$ inside the injectivity radius of p . Moreover we can control r such that d^2 have positive Hessian on $B(p, r)$. More precisely we have the following well theorem:

Theorem 9 (See [35]). *Let M be a Riemannian manifold, consider $p \in M$ and let*

$$B(0, r) := \{v \in T_p M \quad : \quad \|v\| \leq r\}$$

such that the exponential map $\exp_p : B(0, r) \rightarrow M$ is a diffeomorphism over its image. Denote by $B(p, r) = \exp_p(B(0, r))$ the geodesic ball with center p and radius r . Suppose that the sectional curvature K on $B(p, r)$ satisfies the inequality

$$K \leq \mu, \quad \mu \geq 0,$$

and that

$$r < \frac{\pi}{2\sqrt{\mu}} \tag{A.16}$$

if $\mu > 0$. Then the square of the distance to p function $d^2(\cdot, p) : B(p, r) \rightarrow \mathbb{R}$ has positive definite Hessian.

We can use this theorem in order to construct Riemannian metrics on Ω such that $V - \omega$ can be left without damping. For instance, if $(q_1, \dots, q_d) \in \mathbb{R}^d$ is a fixed point, then we can consider the function $d^2(x_1, \dots, x_d) = \sum_{i=1}^d (x_i - q_i)^2$ restricted to Ω and put a Riemannian metric with sufficiently small constant curvature K in $B(q, r) \supset \Omega - \omega$ such that $B(q, r)$ satisfies the conditions of Theorem 9. Then $V - \omega$ can be left without damping.

Example A.1. Denote the hyperbolic plane by $\mathbb{H}^2$. The half plane model of $\mathbb{H}^2$ is given by the set $\mathbb{R}_+^2 = \{(x_1, x_2) \in \mathbb{R}^2; x_2 > 0\}$ endowed with the Riemannian metric $G_{ij} = (1/x_2^2)\delta_{ij}$, where δ_{ij} is the Kroenecker delta function. The geodesics in $\mathbb{H}^2$ are the vertical lines and the semicircles with the origin in the x -axis. Every geodesic is a minimizer, that is, the length of a geodesic that connects x and y is the distance between x and y . Moreover curves that are arbitrarily close to the x -axis (for instance, geodesics in $\mathbb{H}^2$) have infinite length.

Bounded domains Ω in $\mathbb{H}^2$ have compact closures in $\mathbb{H}^2$. From all these facts, it is straightforward that every geodesic does not remain in any bounded domain.

Alternatively we could observe that $\mathbb{H}^2$ has constant Gaussian curvature equal to -1 and that the exponential maps in $\mathbb{H}^2$ are diffeomorphisms $\exp_p : T_p M \rightarrow \mathbb{H}^2$ for every $p \in \mathbb{H}^2$. Now we can use Theorem 9 in order to reach the same conclusion.

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